

➤ IDE

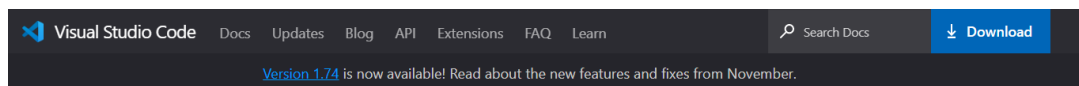
- IDE stands for **Integrated development environment**.
- It is nothing more than an enhanced version of a text editor that helps you write more efficient and nicer code.
- It helps to differentiate different parts of your codes with different colors and notifying you if you are missing some semicolon or bracket at some place by highlighting that area.
- A lot of IDEs are available such as **DEV++ Code Blocks, VS Code**.

➤ Visual Studio Code

- Getting up and running with Visual Studio Code is swift and straightforward.
- VS Code is a free text editor created by Microsoft.
- It is a good editor for JavaScript, HTML and CSS.
- It is a small download so you can install it quickly.
- Additionally, it runs on the Windows, macOS, and Linux operating systems.
- It also supports a large number of extensions that allow the VS Code editor to be adapted to many purposes.

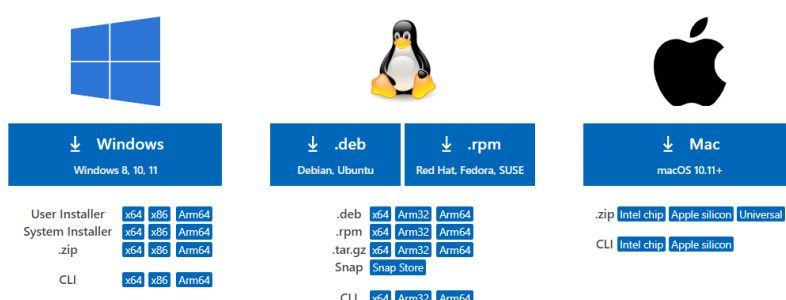
➤ Visual Studio Code Installation

- Visit <https://code.visualstudio.com/download>



Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



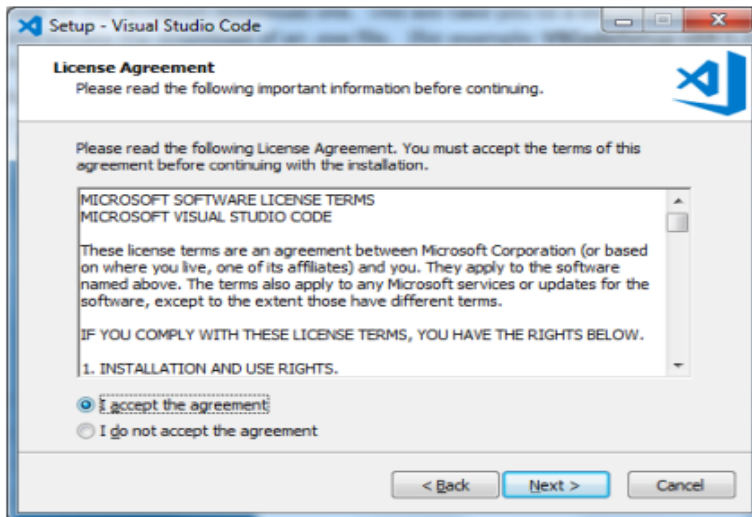
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- Click on the Windows download link. This will take you to a documentation page and also initiate the download of an .exe file.
- Run the exe. Step through the installation wizard as indicated below.

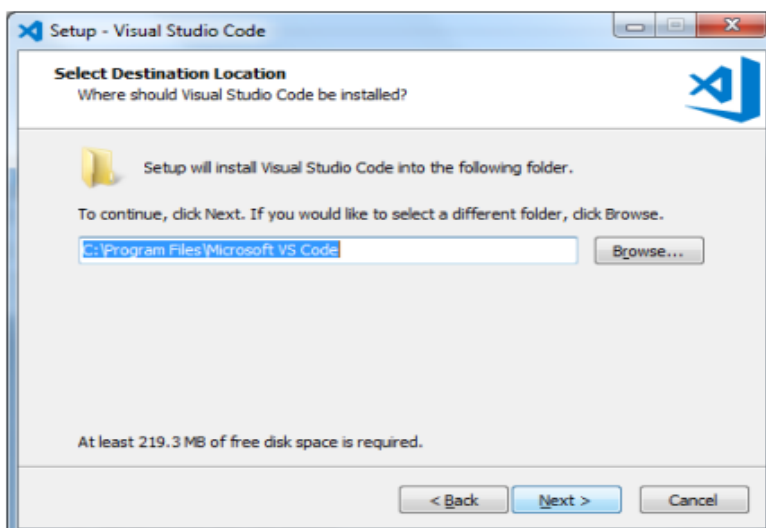


Click Next

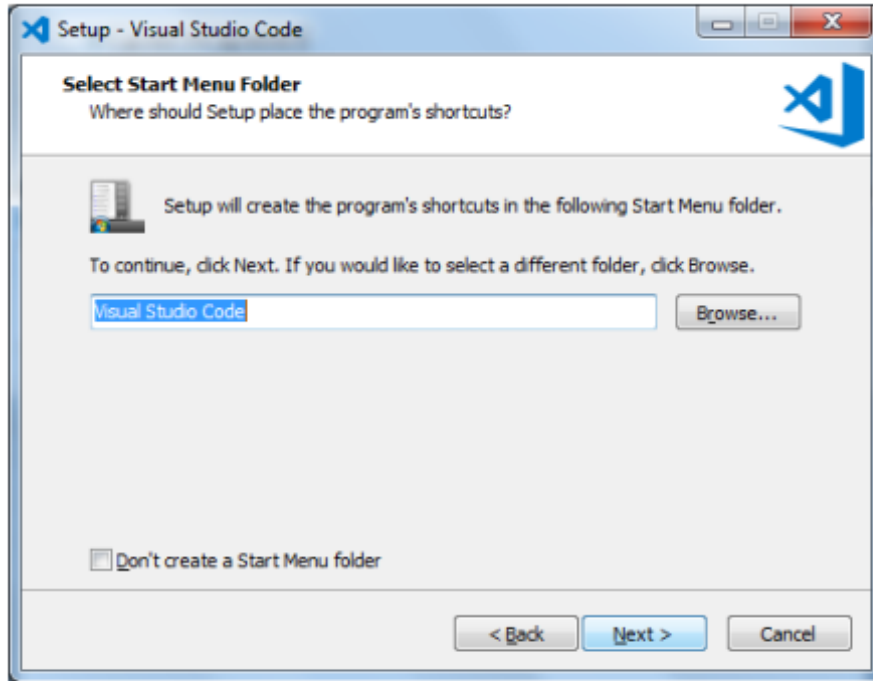
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Accept the license agreement.

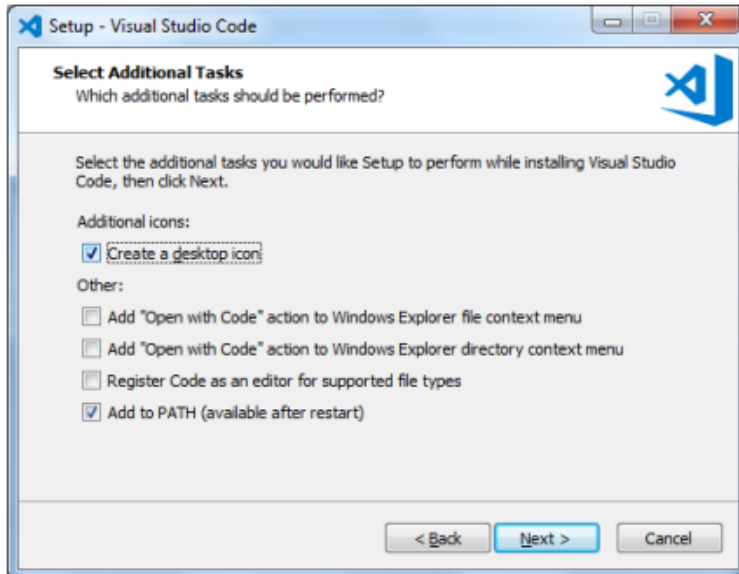


Use the default installation folder. Click Next.

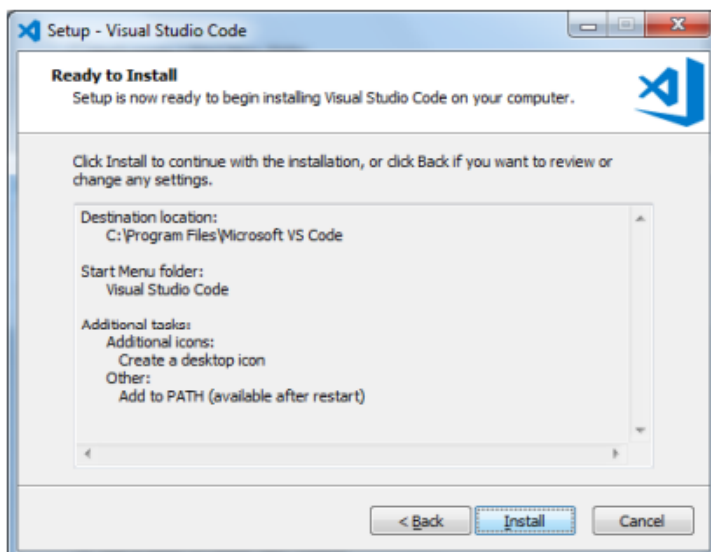


Click Next

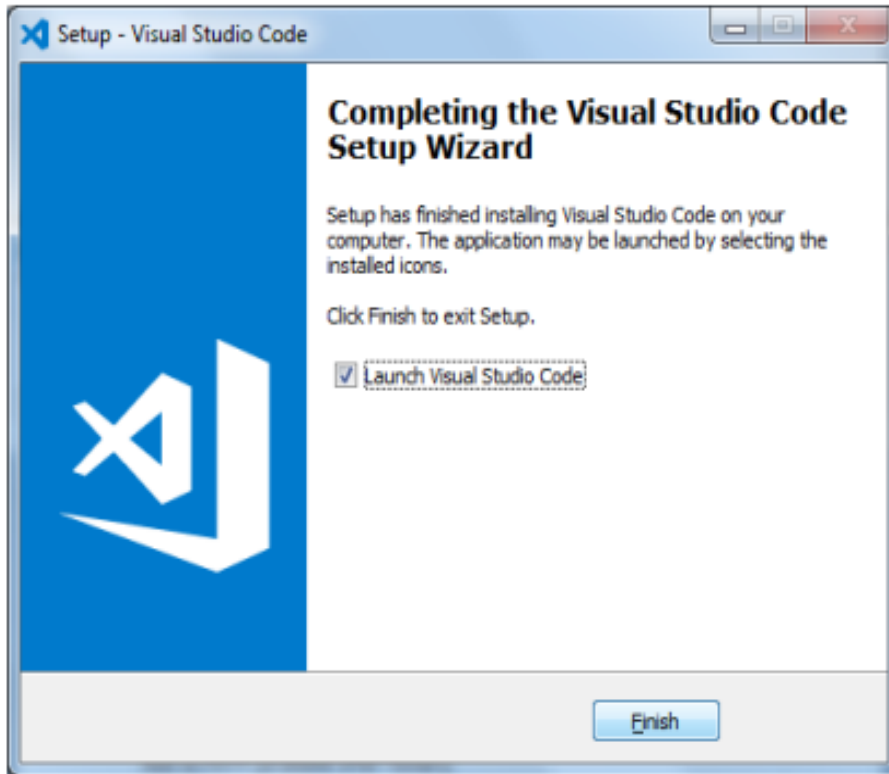
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Creating a desktop shortcut is recommended. You will use VS Code a lot in this course, and you want it to be easy to find. Check the boxes as shown. Click Next

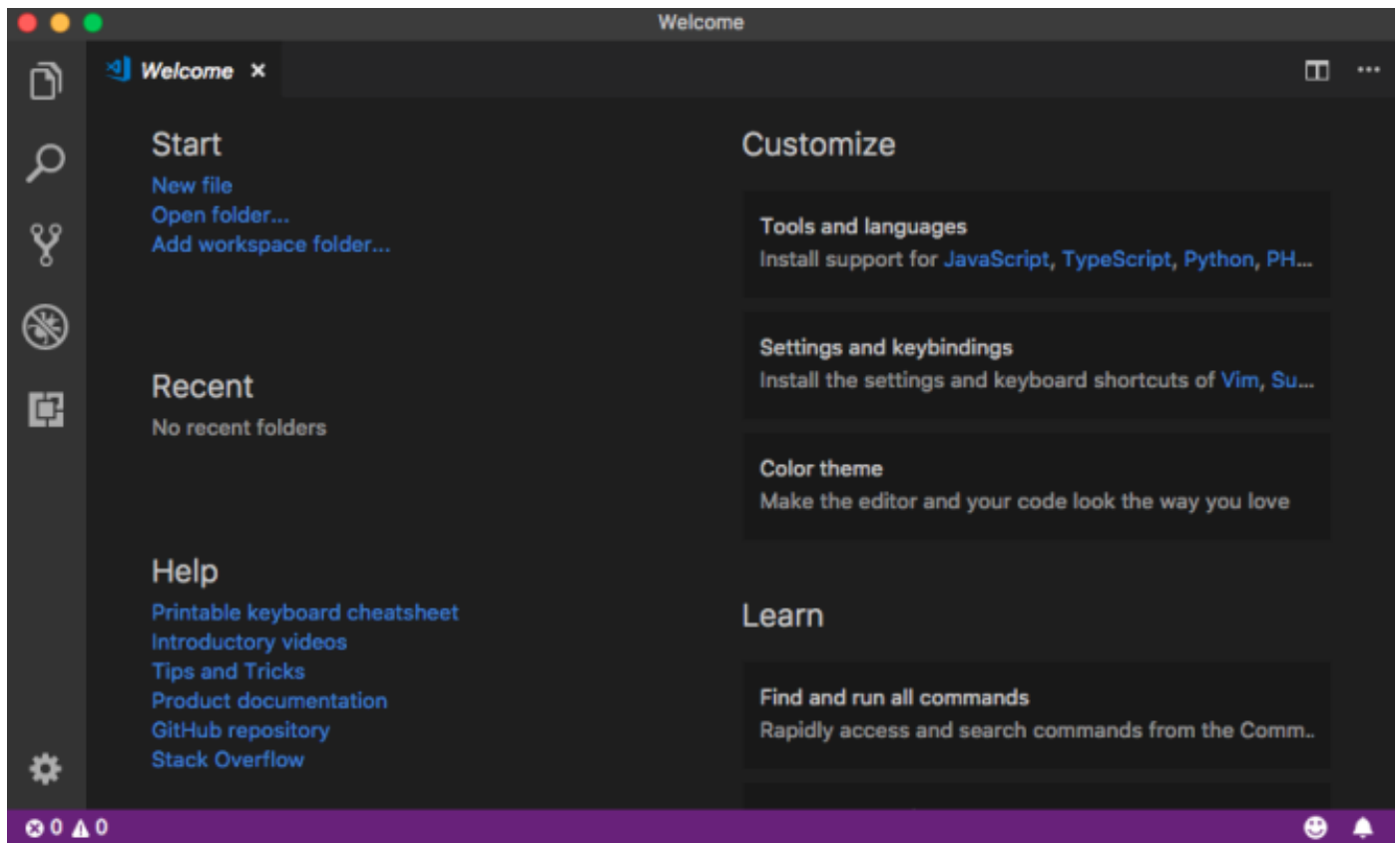


Click Install.



Click Finish.

- VS Code will start, and you will see a screen similar to the following:



- On the left side, you should see a couple of icons. One of them (under the no bugs icon) is the extensions button:

➤ Extensions:

Keyboard Shortcut: `Ctrl+Shift+X`

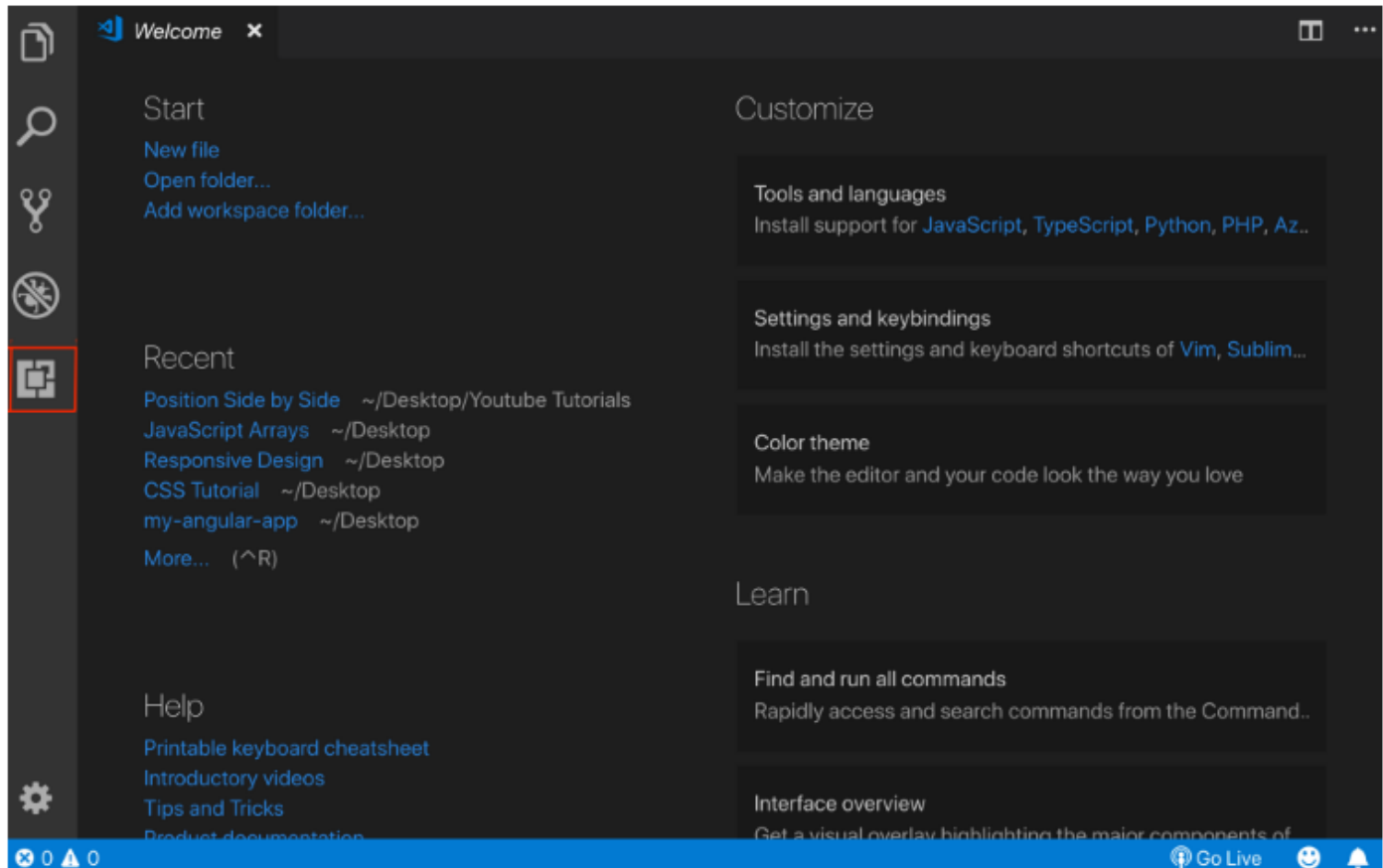
Find extensions:

1. In the VS Code Marketplace.
2. Search inside VS Code in the **Extensions** view.
3. View extension recommendations
4. Community curated extension lists, such as awesome-vscode

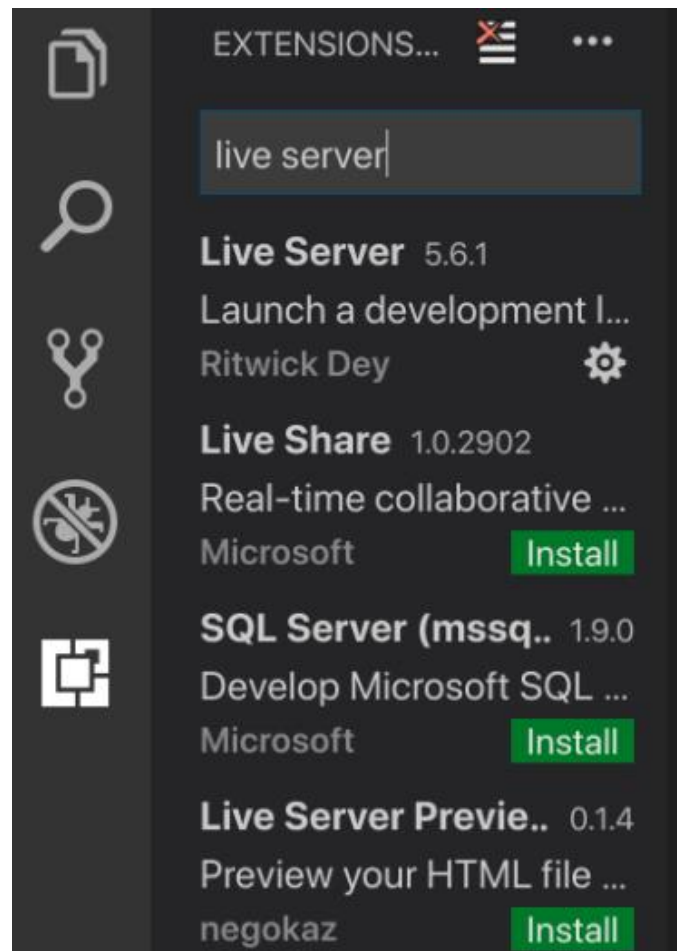
Enable additional languages, themes, debuggers, commands, and more. VS Code's growing community shares their secret sauce to improve workflow.

Installing Extensions:

In the **Extensions** view, you can search via the search bar or click the **More Actions (...)** button to filter and sort by install count.



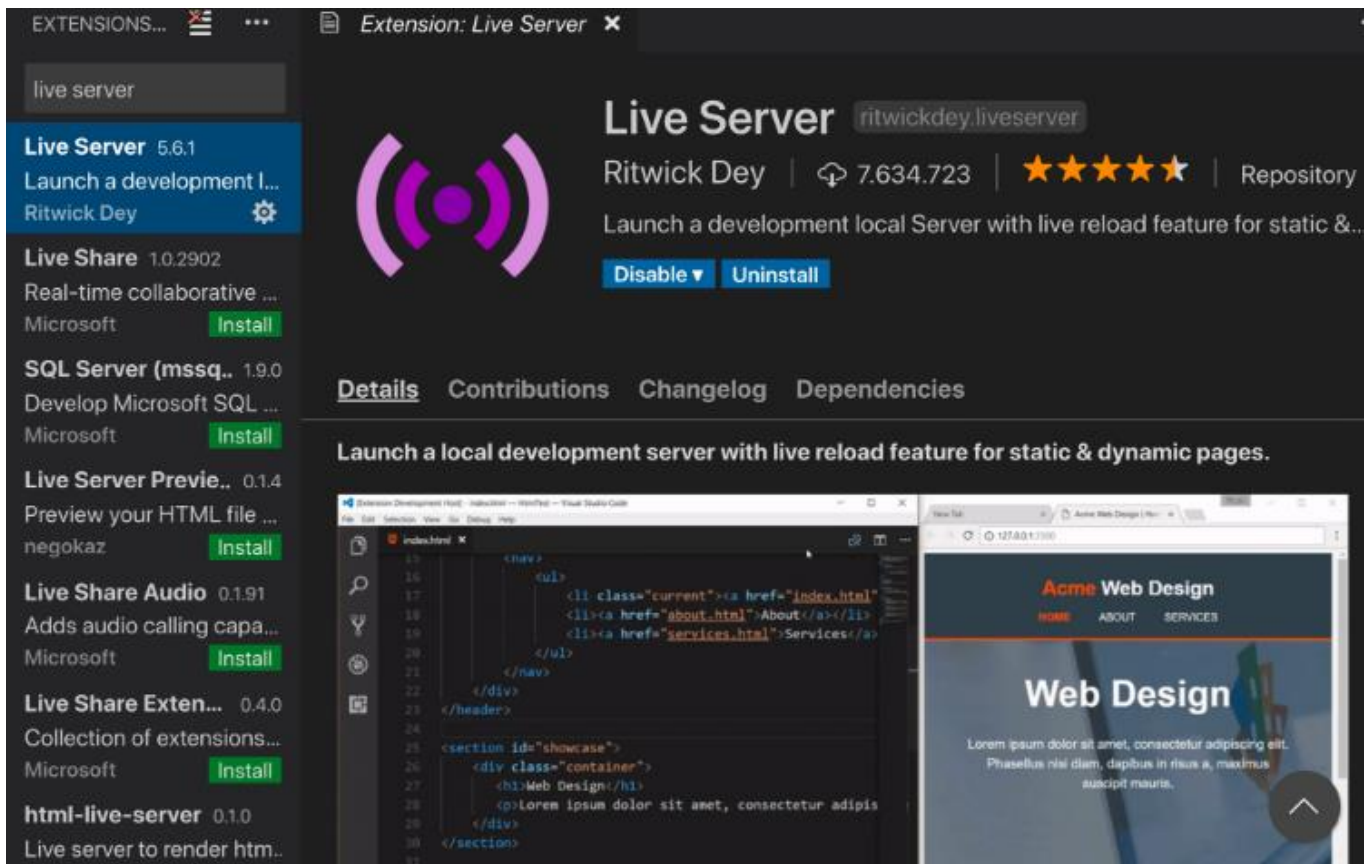
- Once you click on it a search bar will appear. Just type in **“live server”**.



➤ Live Server installation

- You will see many options, so you can choose whichever one works for your system.

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The screenshot shows the VS Code interface with the Live Server extension installed. The left sidebar displays a list of extensions, including Live Server, Live Share, SQL Server, and Live Server Preview. The main panel shows the Live Server extension details, including its icon, name, author (Ritwick Dey), version (5.6.1), and a description: "Launch a development local Server with live reload feature for static &..". The extension is currently disabled, and buttons for "Disable" and "Uninstall" are visible. Below the extension details, there are tabs for "Details", "Contributions", "Changelog", and "Dependencies". The "Details" tab is active, showing a description: "Launch a local development server with live reload feature for static & dynamic pages." Below this, there are two preview images: one showing the VS Code editor with HTML code and another showing a web browser displaying a web page.

➤ Why Live Server

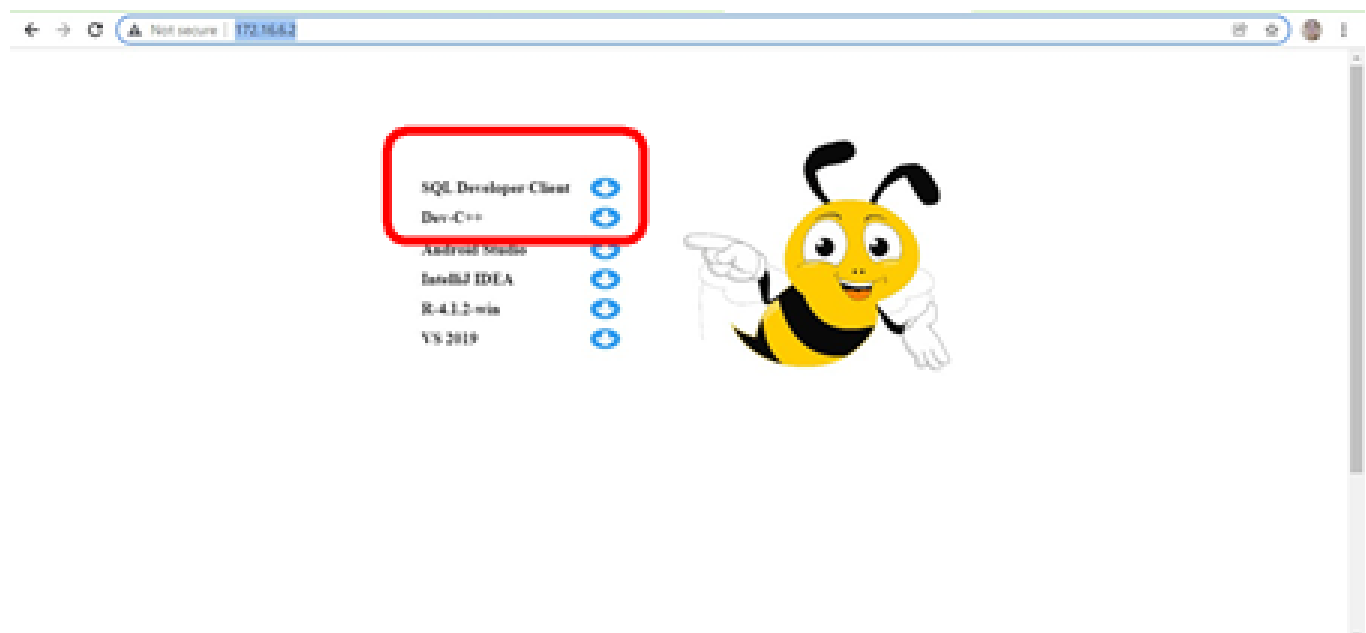
- Normally, when you make a change in your code or write something new, you need to refresh the page manually to see the changes.
- In other words, if you make 100 changes in your code each day, you need to refresh the browser 100 times.
- The live-server extension, however, automates this for you. After installing it, an automated localhost will be able to run in your browser, which you can start with a single button.
- Once you make changes in your code or write something new, after saving it, the browser will auto-refresh itself. Then you will be able to see the changes quickly and automatically.

Objectives

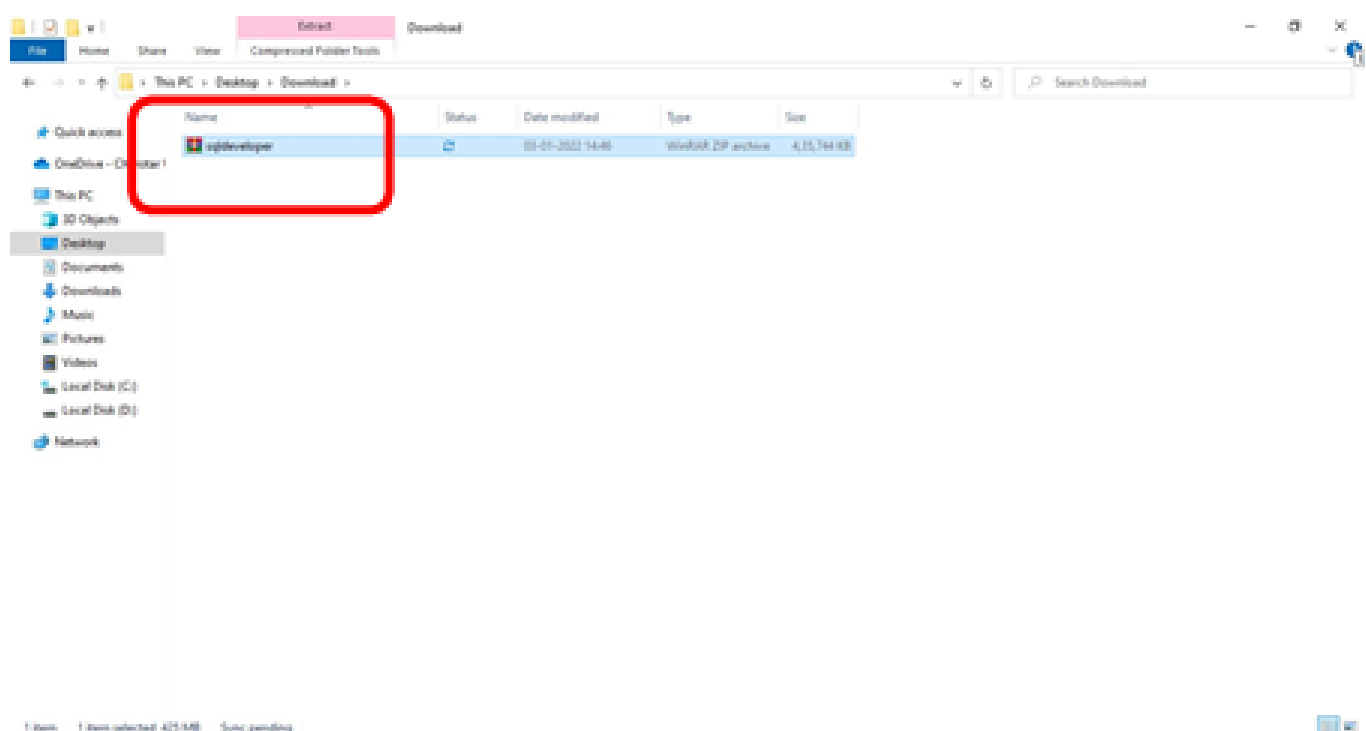
- To make students acquaint with installation of Database Client Software.

Step: 1 Go to <http://172.16.6.2/>

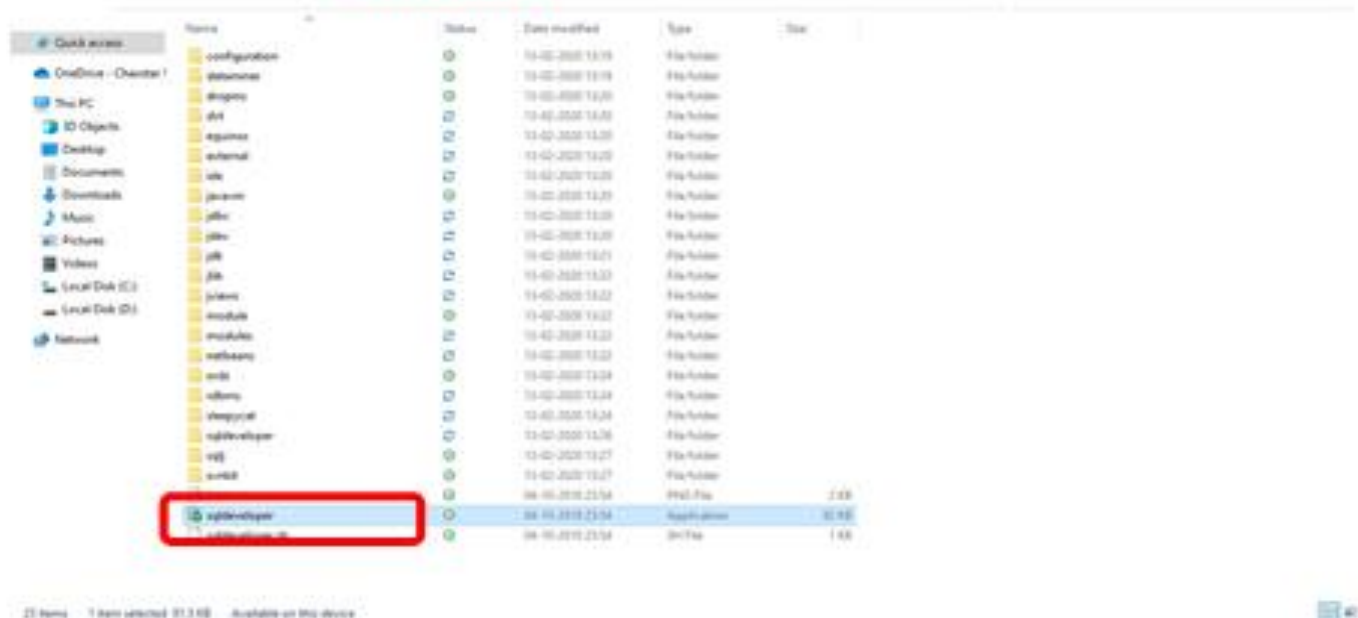
Download SQL Developer Client



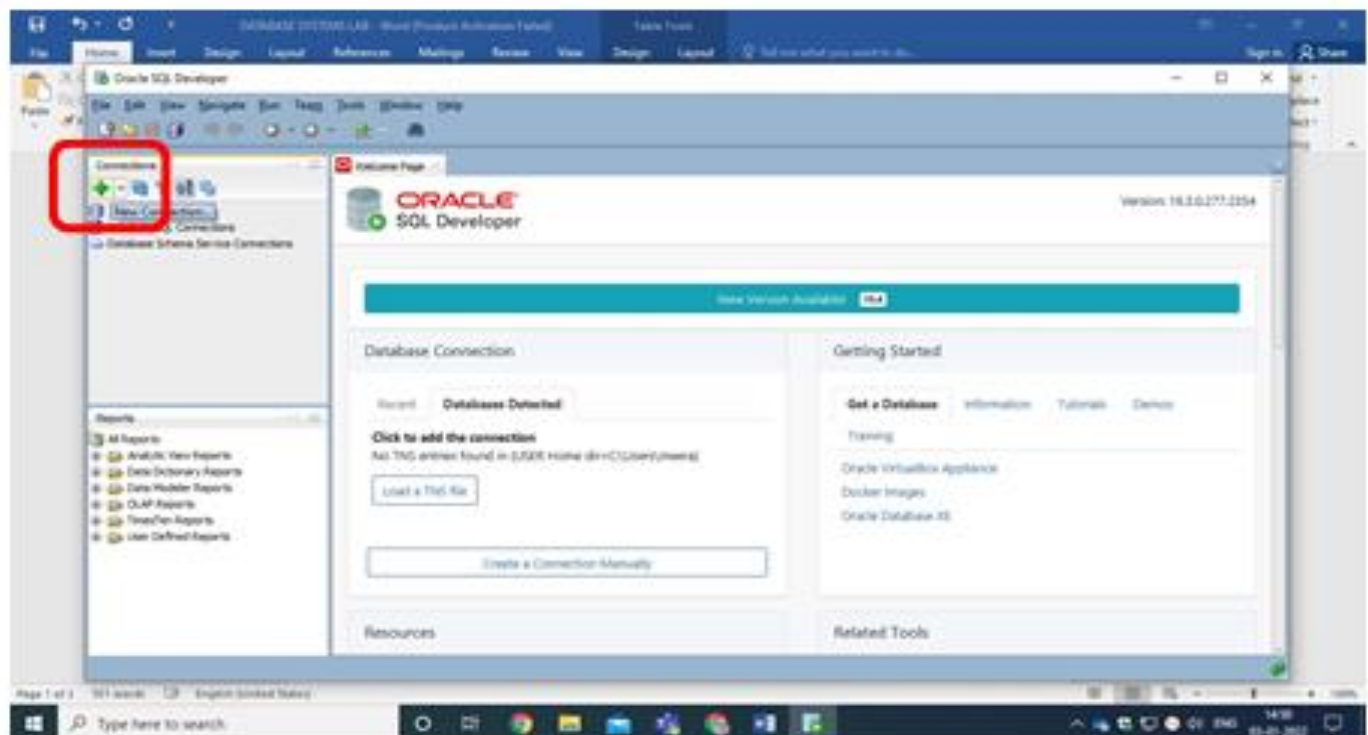
This will download the zip with name sqldeveloper. Unzip it.



Step: 2 Double click on sqldeveloper Application

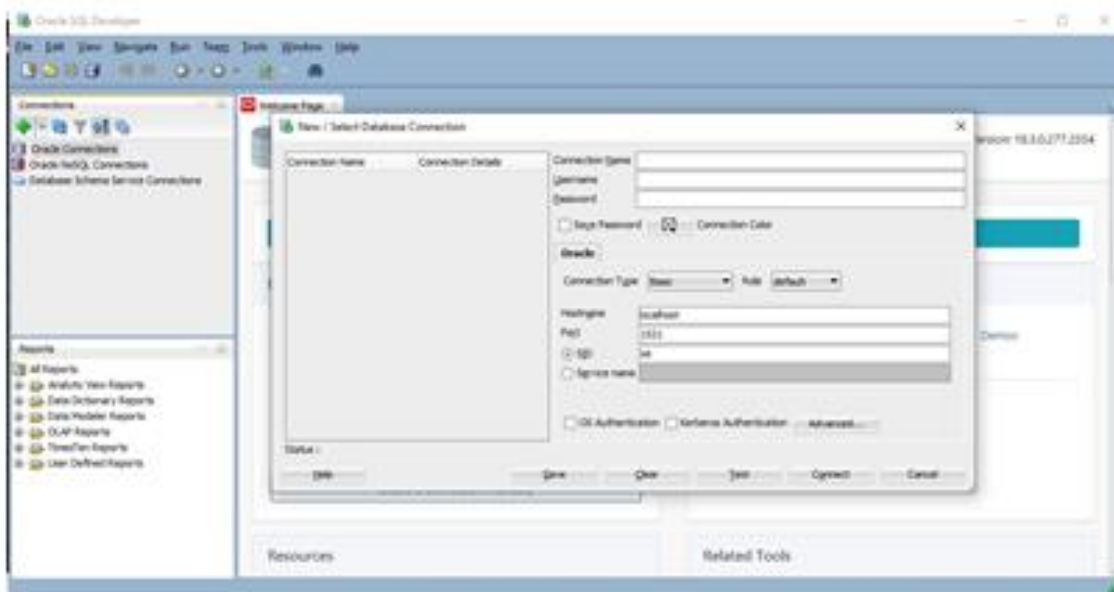


It will open the Oracle SQL Developer.



Step: 3 Click on  Sign, the following screen will appear.

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Step: 4 Give Proper Connection Name for example CMPICA

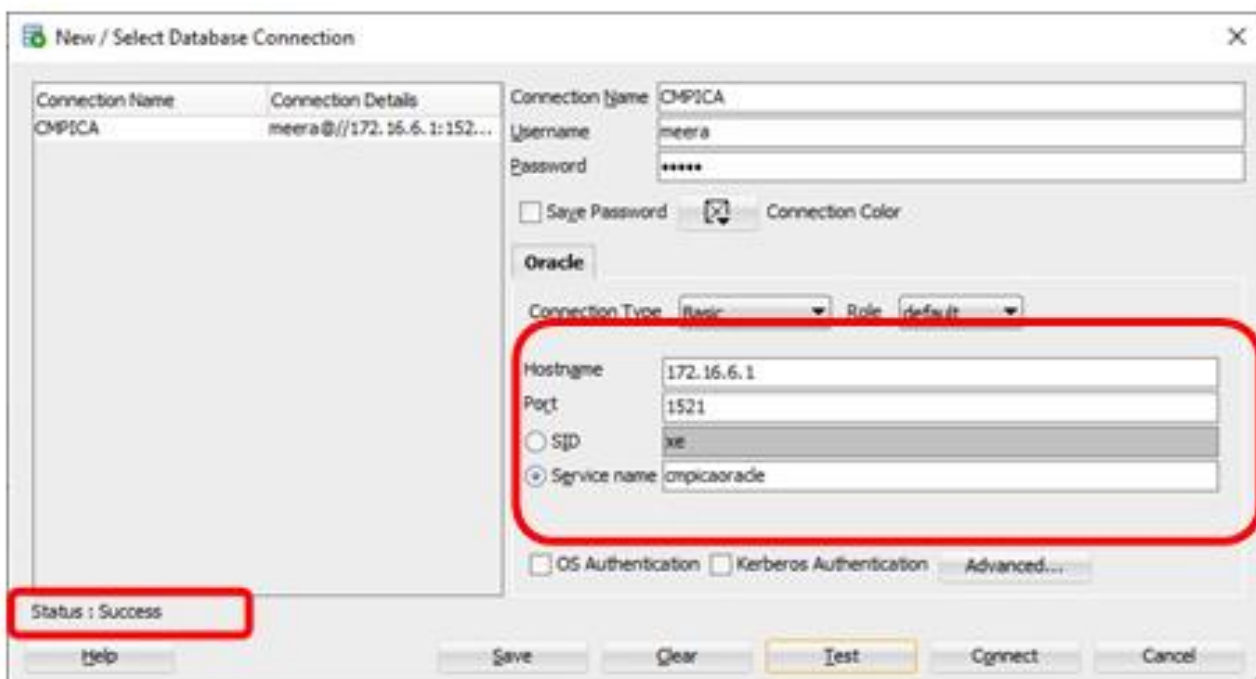
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

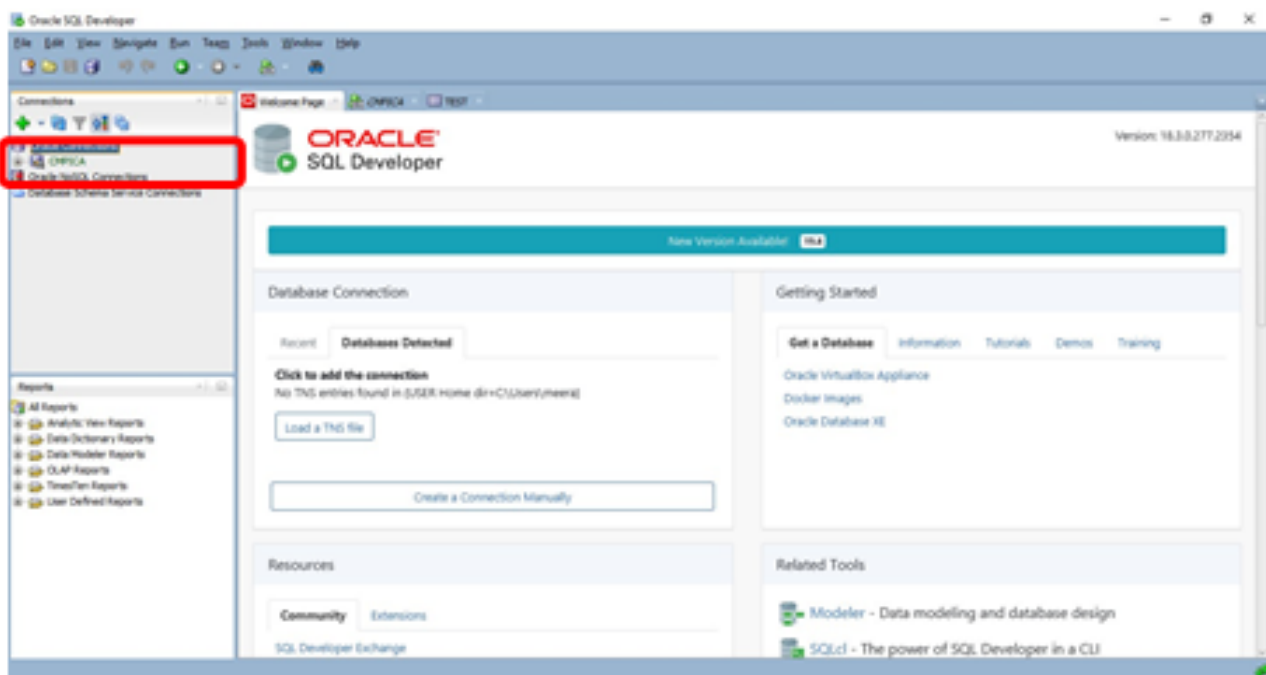
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **SUCCESS** should get displayed.

Now click on Connect Button.

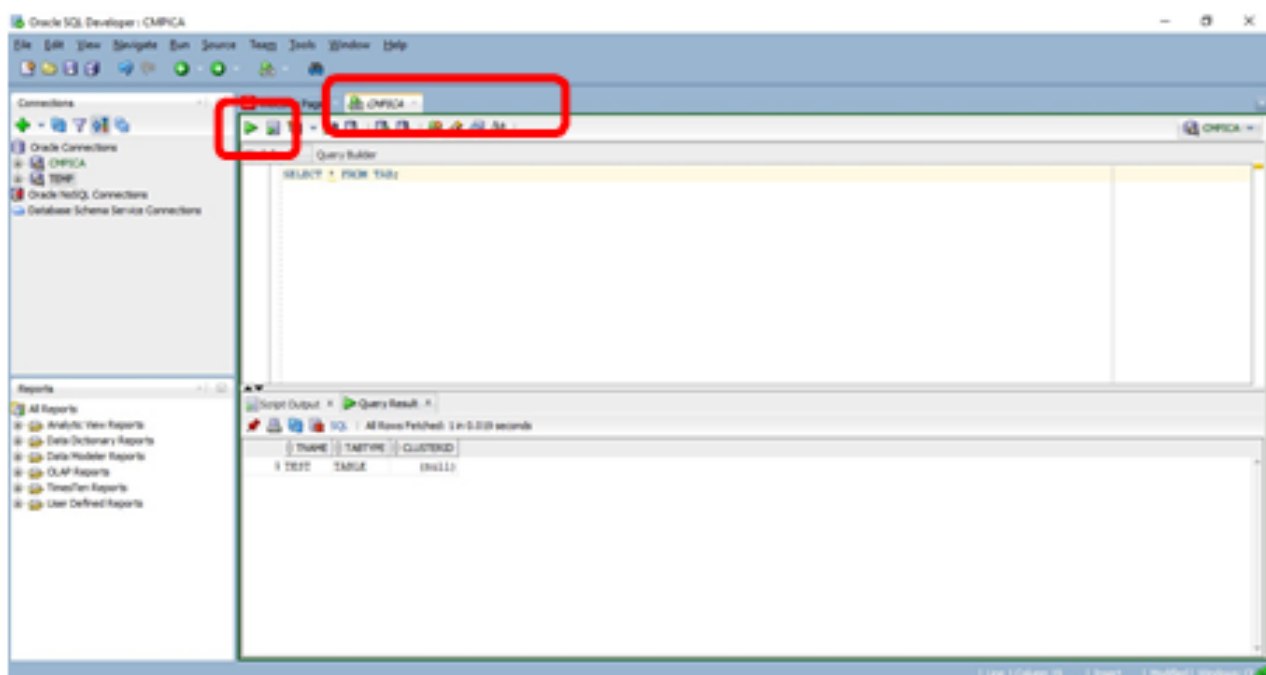


Your newly created connection should get displayed on the left panel.

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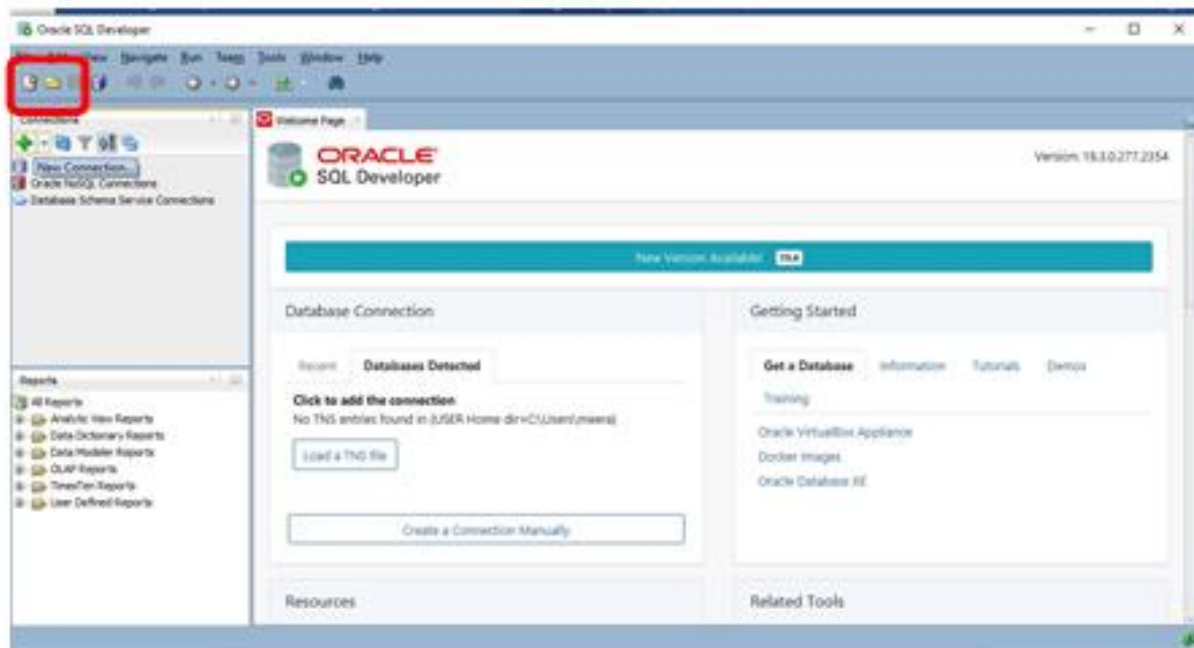
Step 5 On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter.



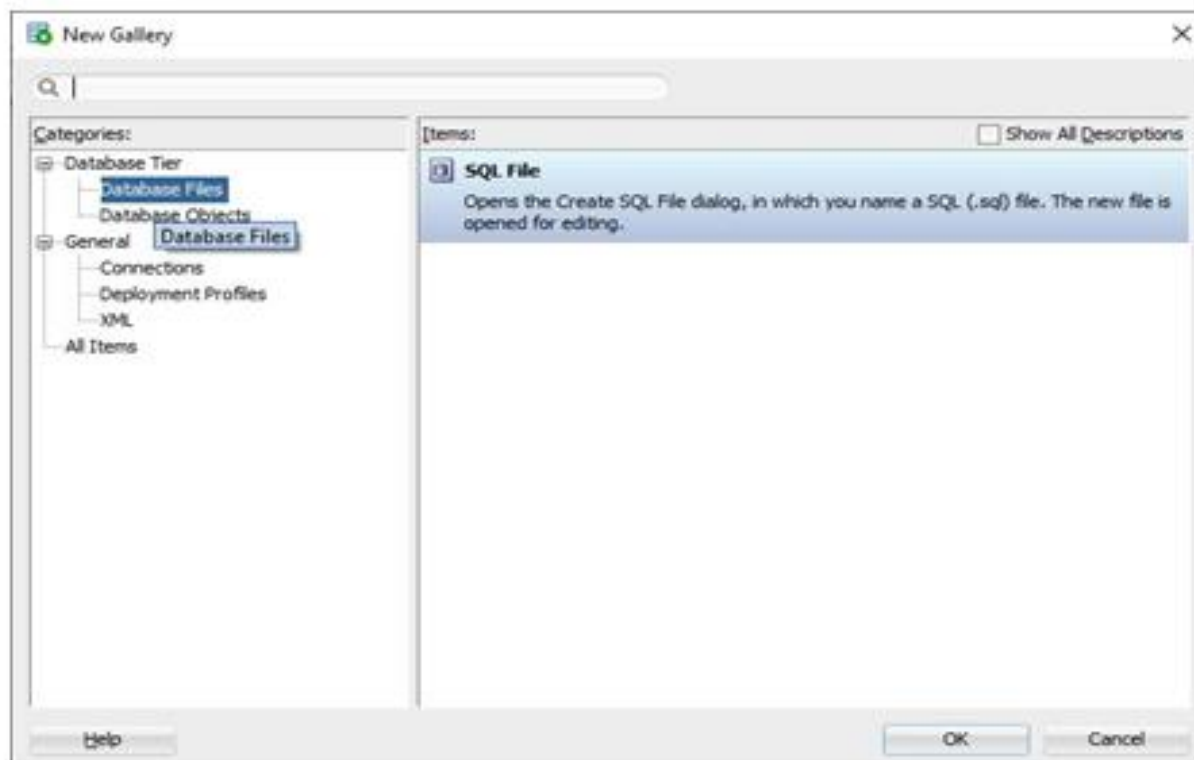
Step 6 IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

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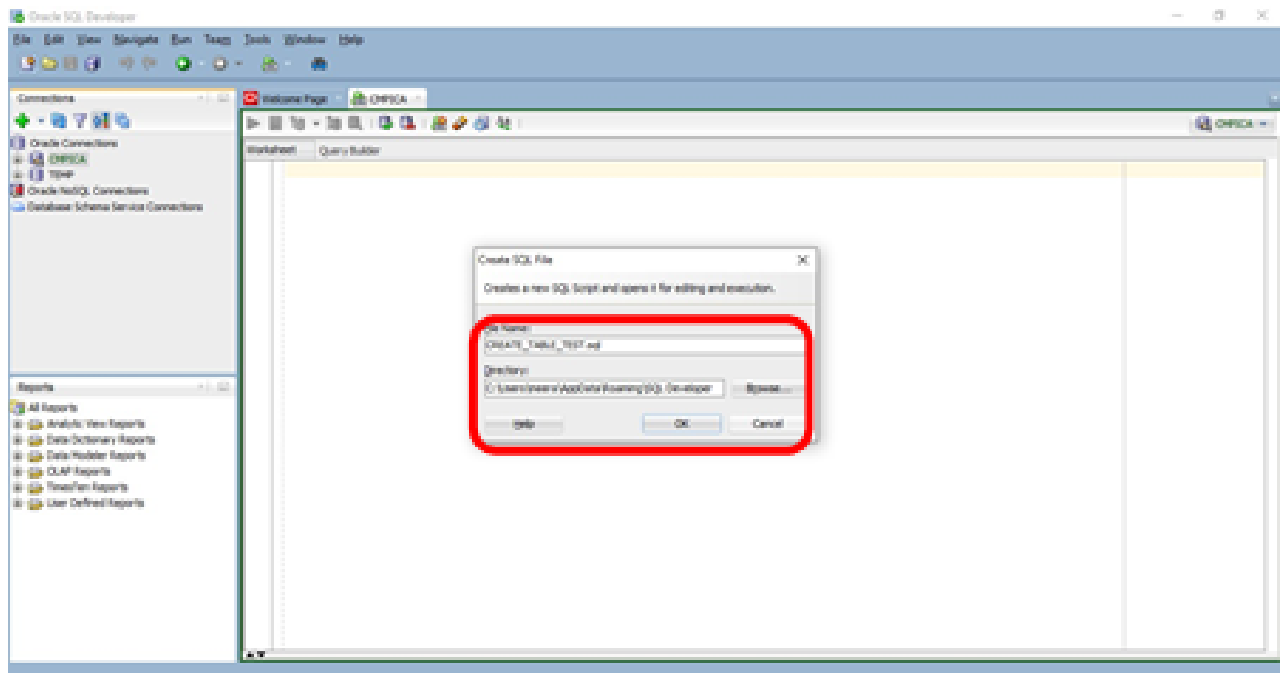


The following screen will appear.



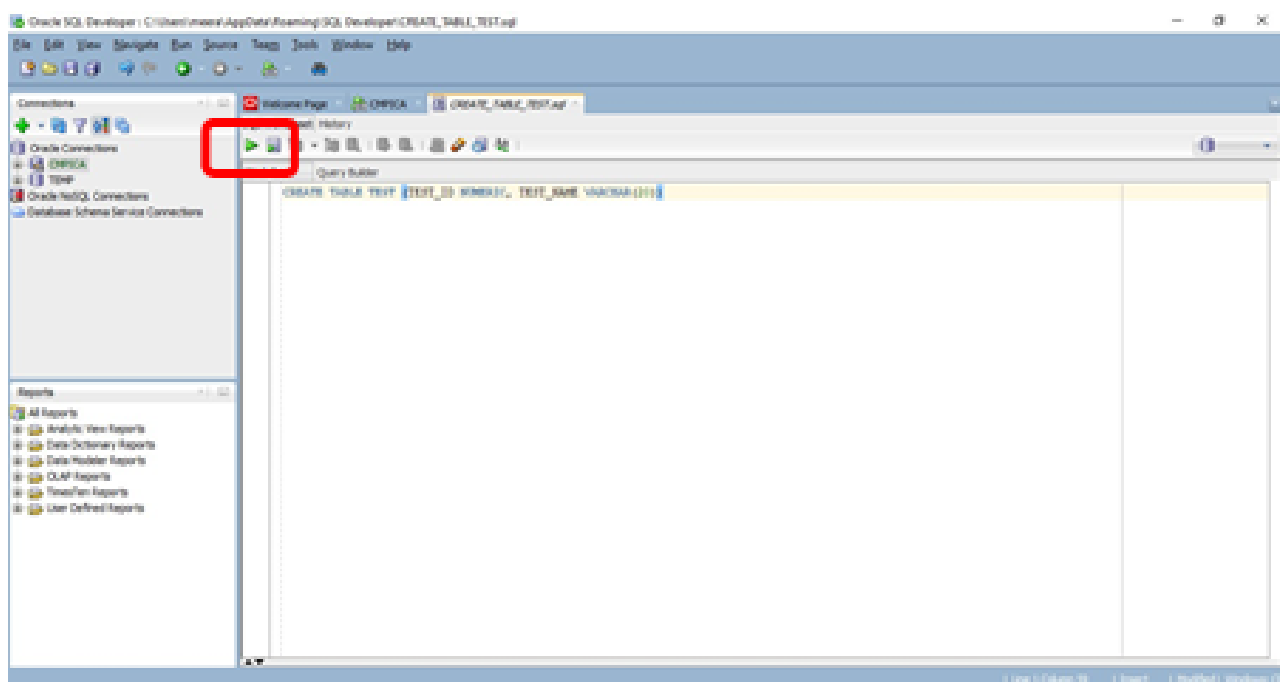
Give proper file name and save to the proper location. [IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]

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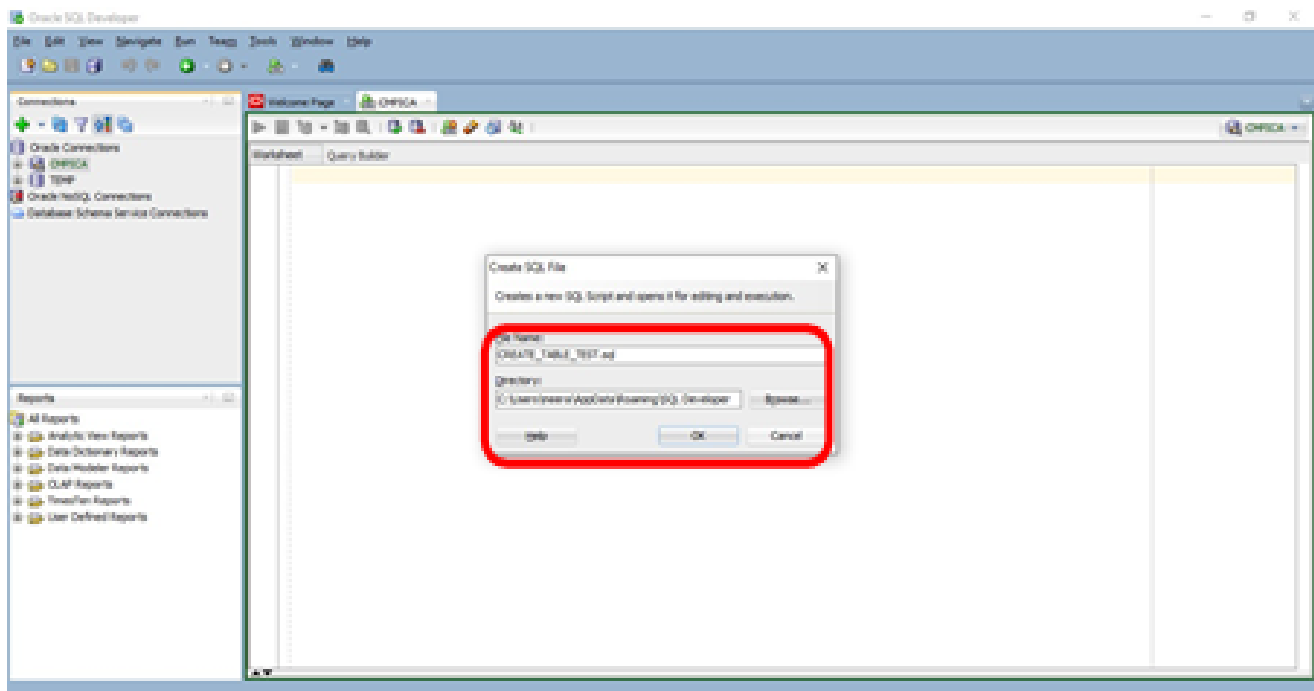
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.



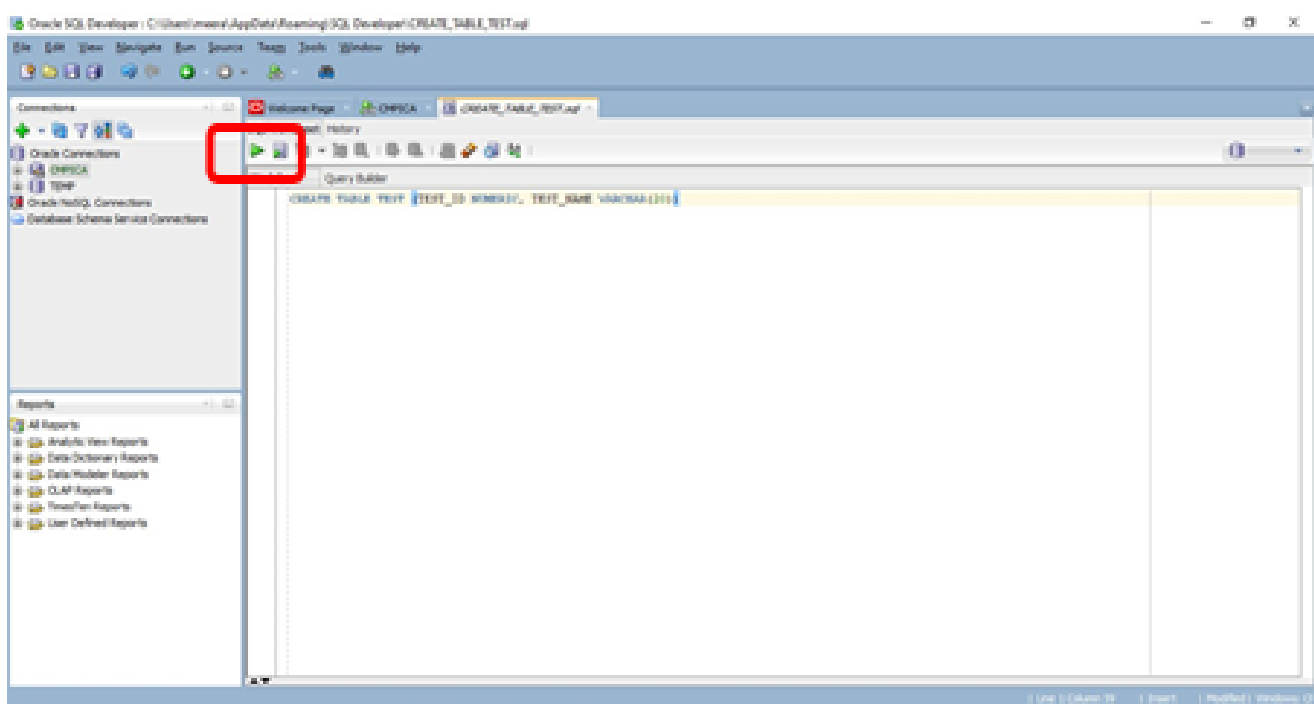
Select the appropriate connection and press OK.

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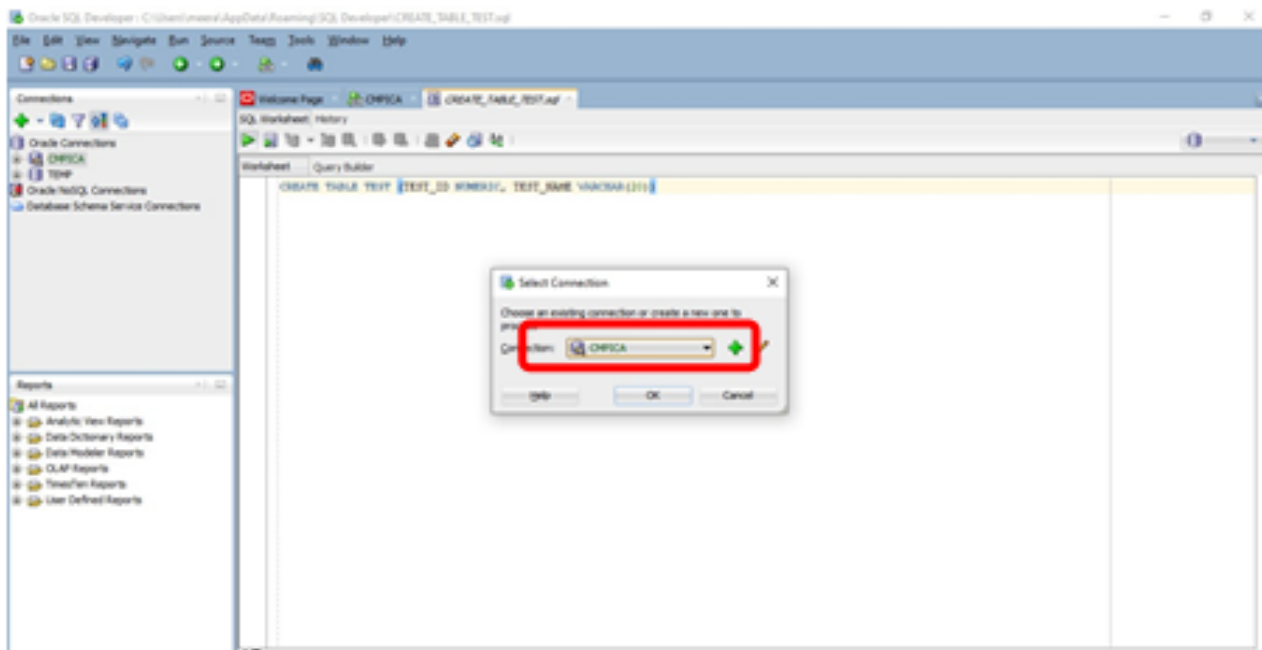
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

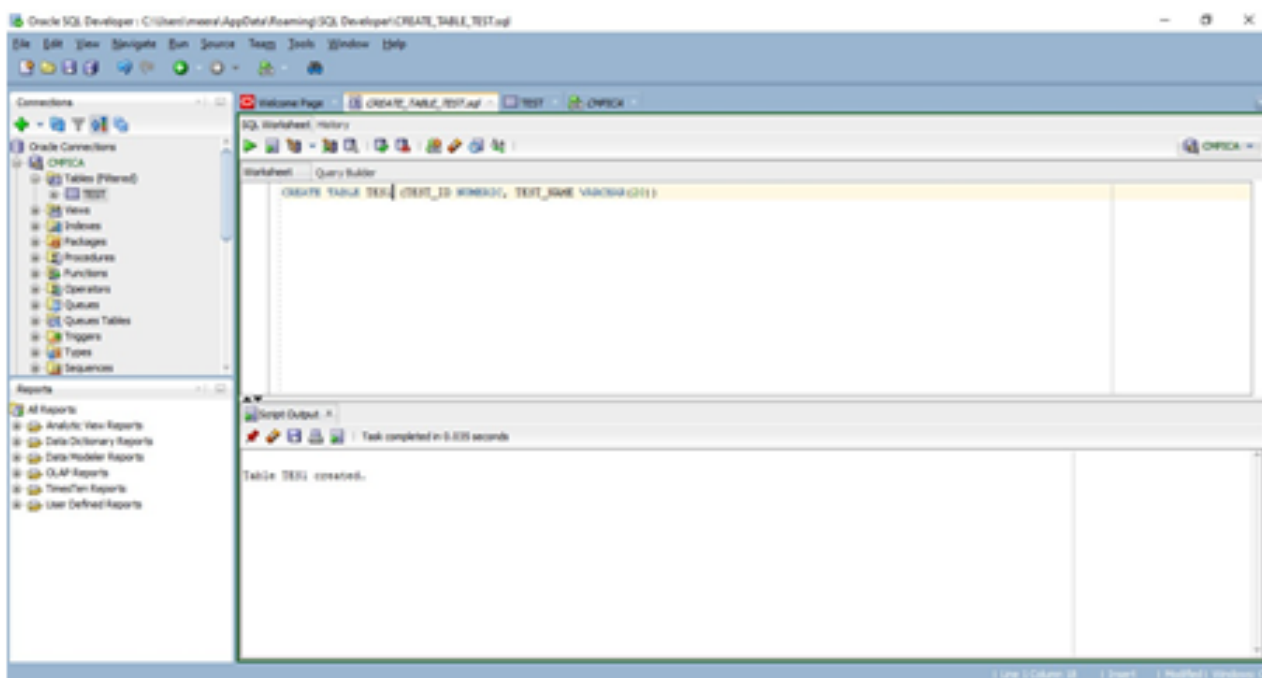


Select the appropriate connection and press OK.

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Observe execution result in bottom panel.



Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.

Visual Studio Code



Visual Studio Code on Windows

Installation

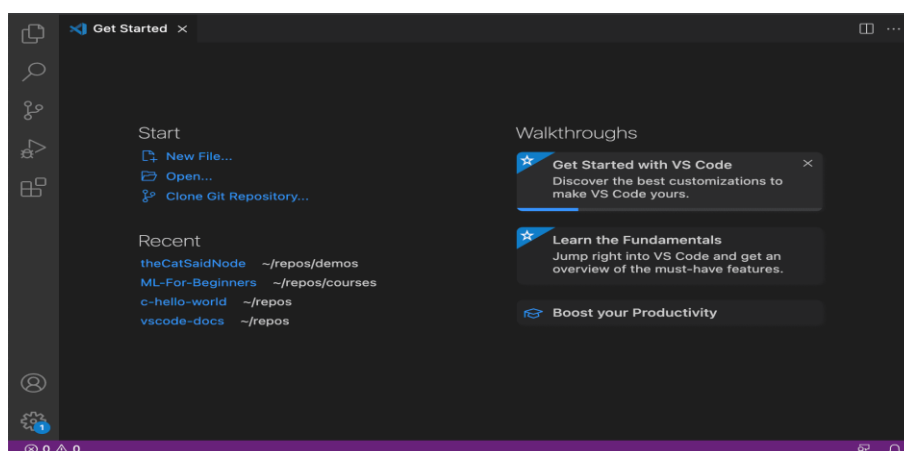
1. Download the [Visual Studio Code installer](#) for Windows.
2. Once it is downloaded, run the installer (VSCodeUserSetup-{version}.exe). This will only take a minute.
3. By default, VS Code is installed under `C:\users\{username}\AppData\Local\Programs\Microsoft VS Code`.

Alternatively, you can also download a [Zip archive](#), extract it and run Code from there.

Introduction:

Visual Studio Code is a lightweight but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support for JavaScript, TypeScript and Node.js and has a rich ecosystem of extensions for other languages (such as C++, C#, Java, Python, PHP, Go) and runtimes (such as .NET and Unity).

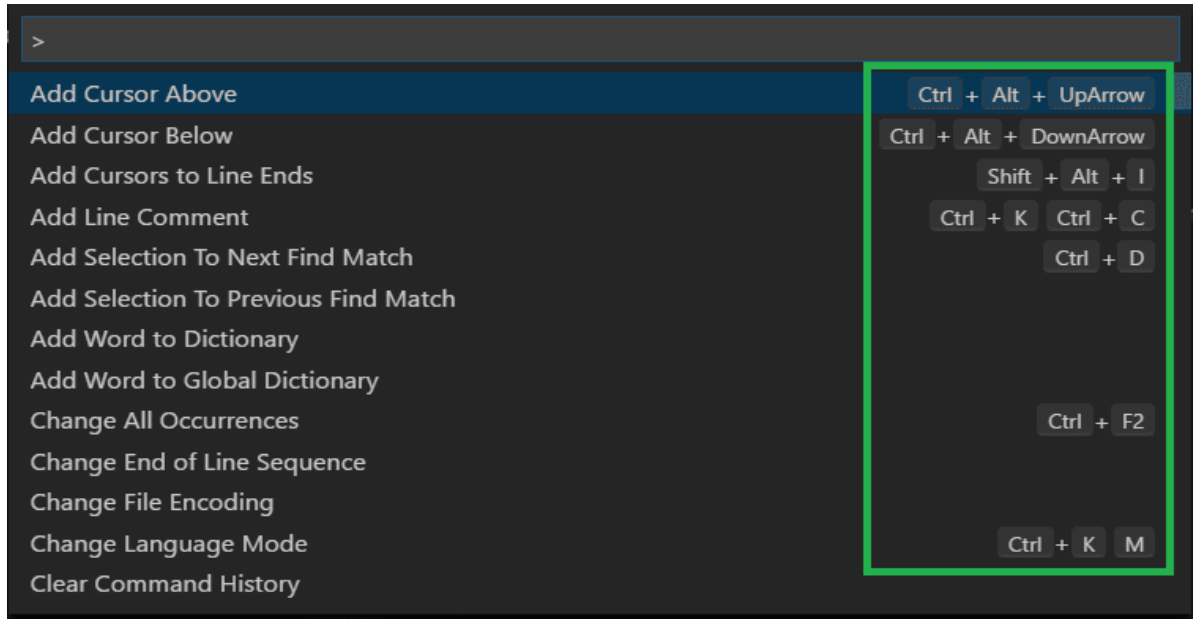
The best way of exploring VS Code hands-on is to open the **Get Started** page. You will get an overview of VS Code's customizations and features. **Help > Get Started**.



Pick a **Walkthrough** for a self-guided tour through the setup steps, features, and deeper customizations that VS Code offers. As you discover and learn, the walkthroughs track your progress.

If you are looking to improve your code editing skills open the **Interactive Editor Playground**. Try out VS Code's code editing features, like multi-cursor editing, IntelliSense, Snippets, Emmet, and many more. **Help > Editor Playground**.

Default keyboard shortcuts



- Create a new file.
 - **File > New File** (Ctrl+N)
- Save a file.
 - **File > Save** (Ctrl+S)
- Open the Command Palette.
 - **View > Command Palette...** (Ctrl+Shift+P)
- See an overview of the user interface.
 - **Help: User Interface Overview** in the Command Palette.
- Install support for your favorite programming language from the Extensions view.
 - **View > Extensions** (Ctrl+Shift+X)
- Change your keyboard shortcuts and easily migrate from other editors using keymap extensions.
 - **File > Preferences > Migrate Keyboard Shortcuts from...**
 - **Code > Preferences > Migrate Keyboard Shortcuts from...** on macOS
- Open the Keyboard Shortcuts editor.
 - **File > Preferences > Keyboard Shortcuts** (Ctrl+K Ctrl+S)
 - **Code > Preferences > Keyboard Shortcuts** on macOS
- Printable keyboard cheat sheet.
 - **Help > Keyboard Shortcuts Reference** (Ctrl+K Ctrl+R)
- Customize your editor with color themes.
 - **File > Preferences > Color Theme** (Ctrl+K Ctrl+T)
 - **Code > Preferences > Color Theme** on macOS
- Show all commands.
 - **Help > Show All Commands** (Ctrl+Shift+P)
- Explore VS Code features in the **Interactive Editor Playground**.
 - **Help > Editor Playground**

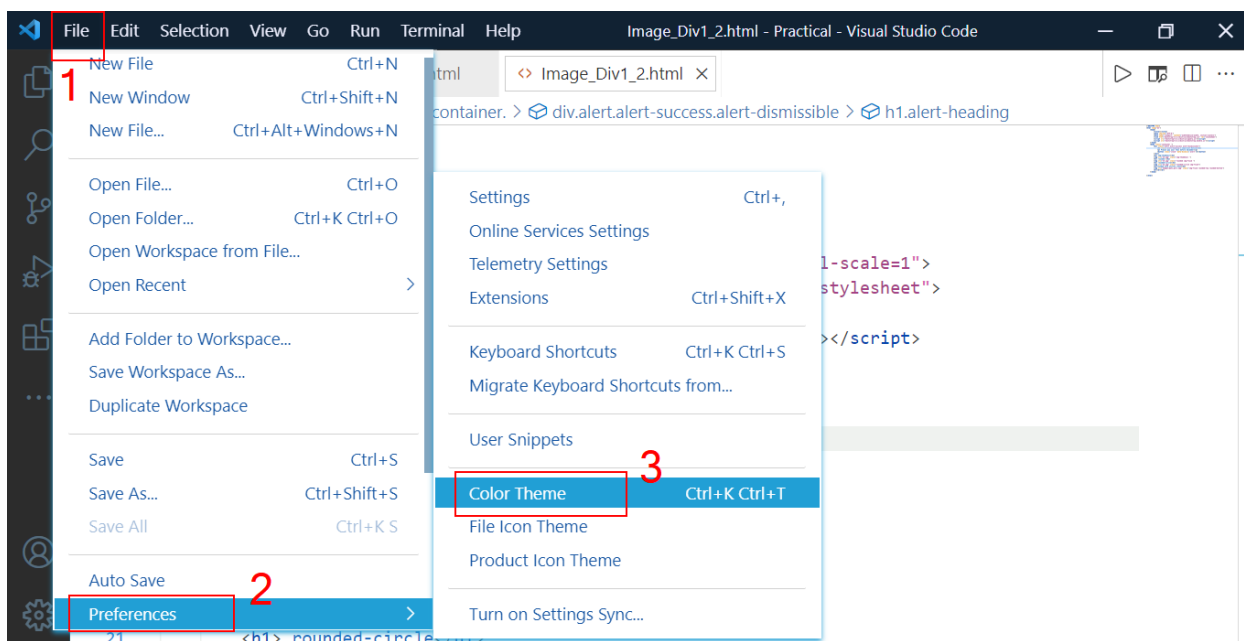
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Customization

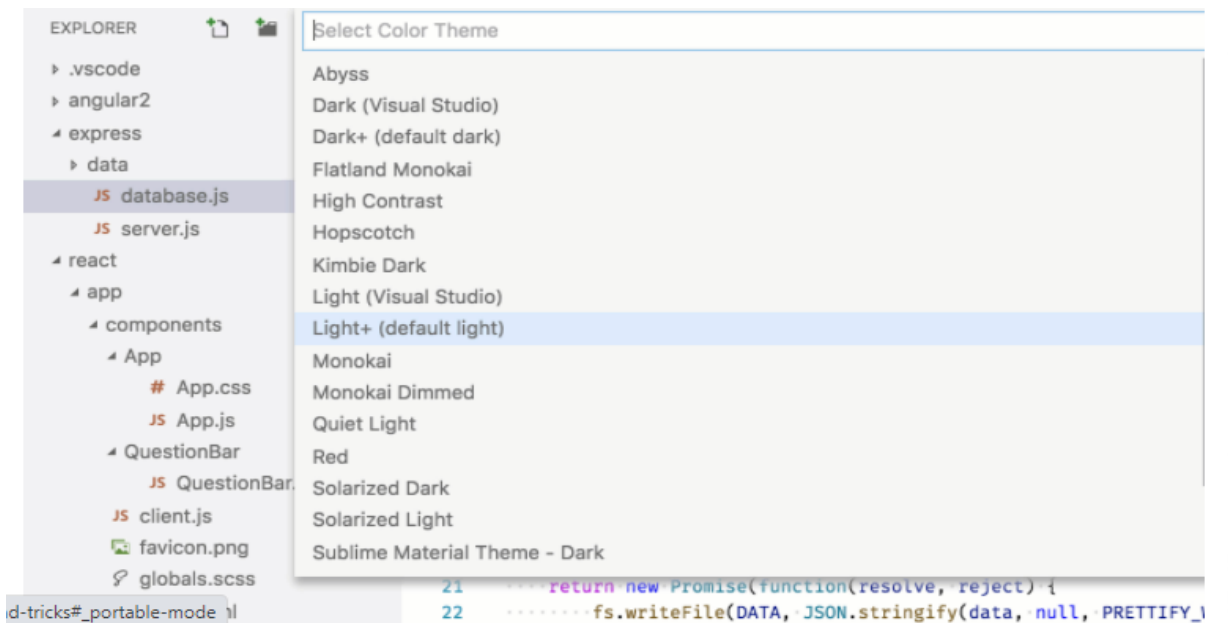
There are many things you can do to customize VS Code.

- Change your theme
- Change your keyboard shortcuts
- Tune your settings
- Add JSON validation
- Create snippets
- Install extensions

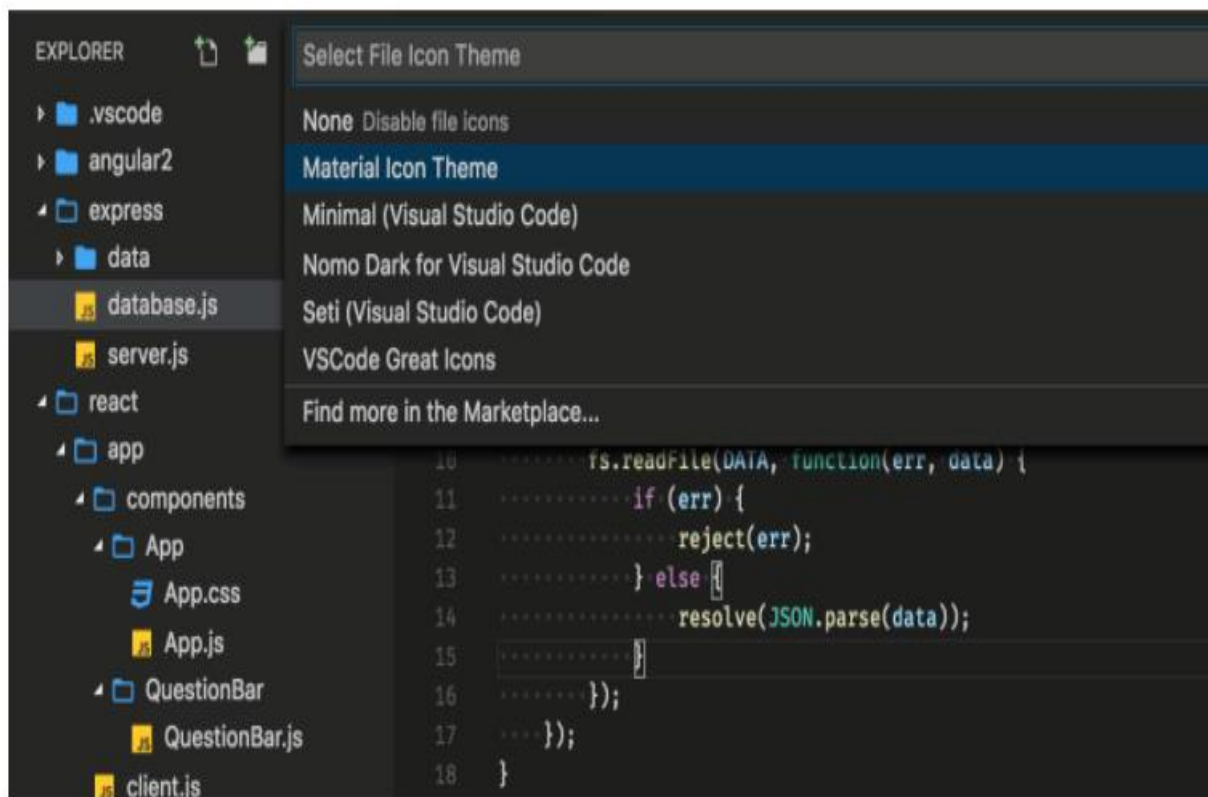
Change Theme



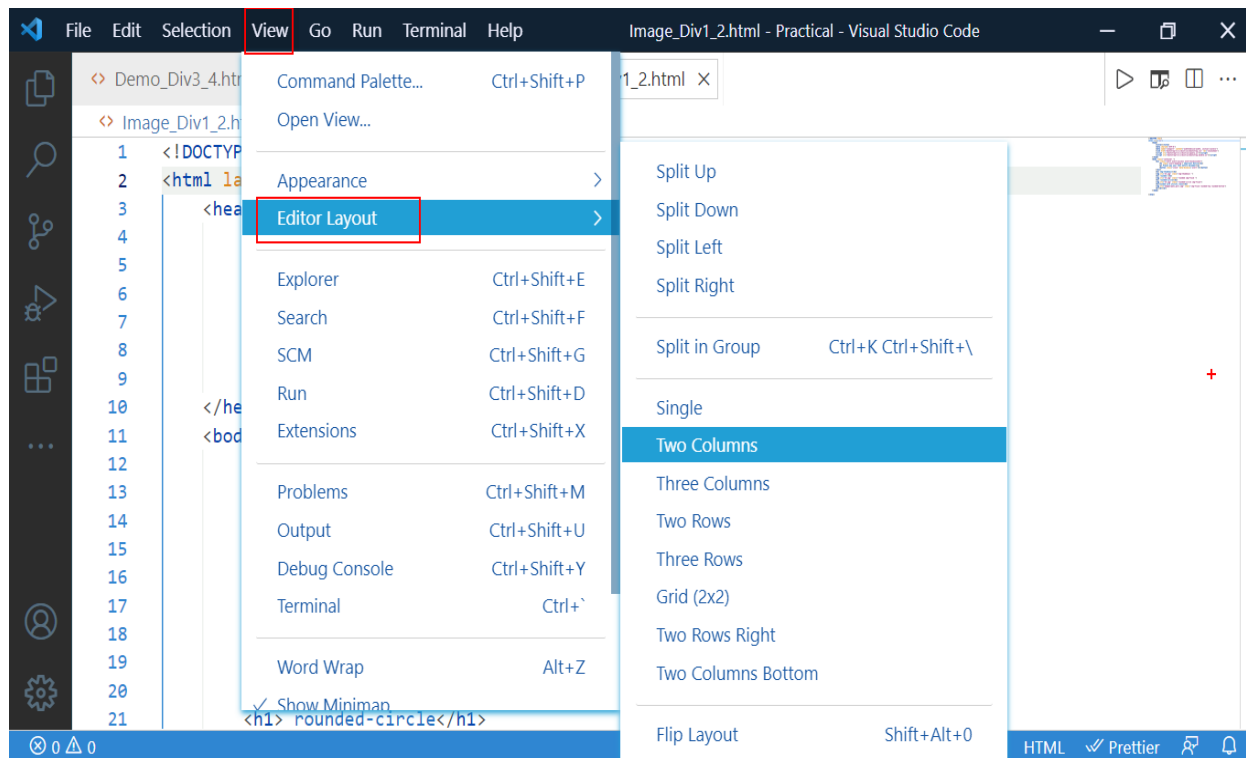
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Additionally, you can install and change your File Icon themes.



Change Editor Layout: (View -> Editor Layout)



Extensions:

Keyboard Shortcut: **Ctrl+Shift+X**

Find extensions:

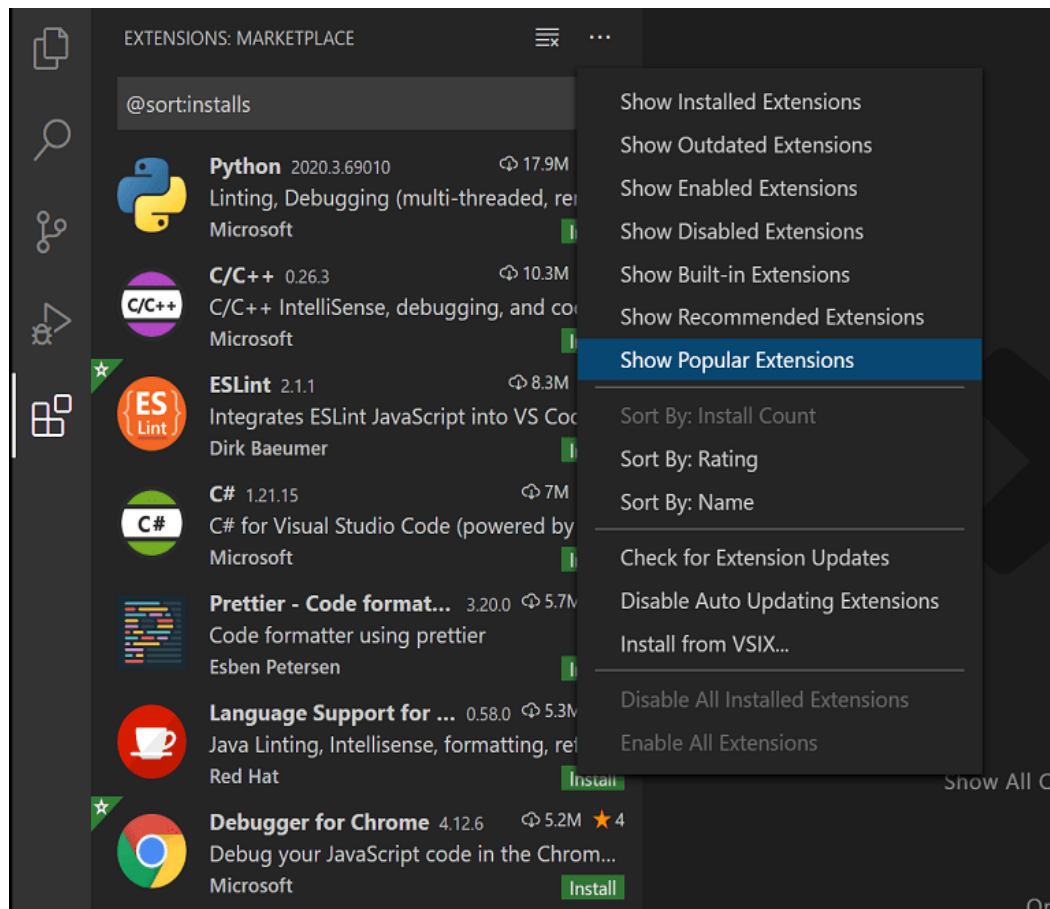
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4. Community curated extension lists, such as awesome-vscode

Enable additional languages, themes, debuggers, commands, and more. VS Code's growing community shares their secret sauce to improve workflow.

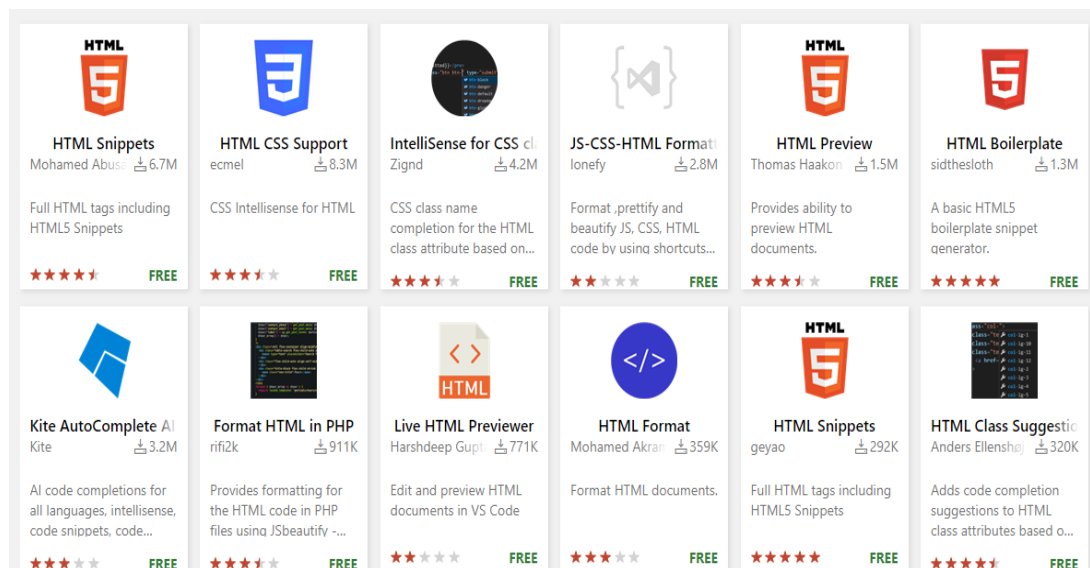
Installing Extensions:

In the **Extensions** view, you can search via the search bar or click the **More Actions (...)** button to filter and sort by install count.

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HTML,CSS Extensions:



Bootstrap Extensions:

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 Bootstrap 4, Font awe Ashok Koyi ⬇ 1.9M Bootstrap 4 snippets based on documentation + Font awesome 4 + Fo... ★★★★★ FREE	 Bootstrap 3 Snippets William Whitehe ⬇ 672K Bootstrap 3 Snippets for VS Code ★★★★★ FREE	 Bootstrap v4 Snippets Zaczero ⬇ 435K Bootstrap v4 Snippets for VS Code ★★★★★ FREE	 Bootstrap 4 CDN Snip Simen Daehlin ⬇ 183K dehlin.dev This will get you up and running to get a boilerplate of bootstrap... ★★★★★ FREE	 Angular UI Bootstrap HerrHerrmann ⬇ 165K Angular UI Bootstrap Snippets for VS Code (HTML and JS) ★★★★★ FREE	 Bootstrap 5 Quick Sni Anbuselvan Rock ⬇ 46.8K Collection of Bootstrap 5 Snippets that will make your life easier. ★★★★★ FREE
 bootstrap-snippets-vs bhavana ⬇ 54.6K ★★★★★ FREE	 The Powerful Bootstrap Crozzo ⬇ 52.1K The Powerful Bootstrap Snippet ★★★★★ FREE	 Bootstrap 5 & Font Aw HansUXdev ⬇ 71.7K Code snippets for Bootstrap 5 ★★★★★ FREE	 Bootstrap extension p walkme ⬇ 37K The most common extensions for Bootstrap in one pack. This... ★★★★★ FREE	 Bootstrap Product Ico RubenVerg ⬇ 23.6K Adds Bootstrap Icons as product icons for Code. ★★★★★ FREE	 Bootstrap 5 Snippets Coder Foundry ⬇ 22.7K A collection of code snippets using Bootstrap 5 created by the staff at... ★★★★★ FREE

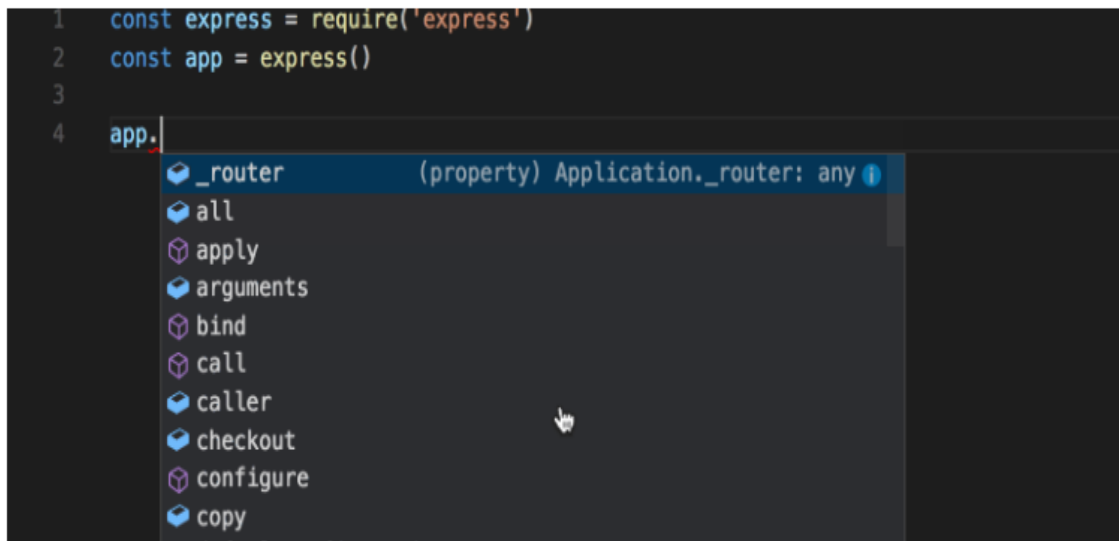
AngularJS Extensions:

 AngularJs 1.x Code Sn alexandersage ⬇ 91.2K AngularJs 1.x Code Snippets ★★★★★ FREE	 AngularJS 1.5+ TypeSc Jimmy Chandra ⬇ 19.6K AngularJS 1.5+ TypeScript Snippets ★★★★★ FREE	 AngularJS Code Snipp NicholasHsiang ⬇ 9.8K Code snippets for AngularJS ★★★★★ FREE	 AngularJS Componen Oleksandr Hutso ⬇ 14.4K This extension helps to generate basic component structure wi... ★★★★★ FREE	 AngularJS Template A Michael Isom ⬇ 13.2K Autocomplete AngularJS controller properties and methods from an html... ★★★★★ FREE	 AngularJS extension p walkme ⬇ 8.8K The most common extensions for Angular JS in one pack. This... ★★★★★ FREE
 AngularJs ngDoc Geni Luis Gonzalez ⬇ 10.1K Generate ngDocs for AngularJS modules, components, directives,... ★★★★★ FREE	 angularjs-reviewcode LocNguyen ⬇ 67 AngularJS (Angular 1) Review Code ★★★★★ FREE	 Angular 1.x snippets f lperdomo ⬇ 33.9K Angular 1.x snippets for Visual Studio Code using John Papa AngularJS stv... ★★★★★ FREE	 AngularDoc for Visual AngularDocio ⬇ 226K Angular development tooling ★★★★★ FREE	 Angular UI Bootstrap HerrHerrmann ⬇ 165K Angular UI Bootstrap Snippets for VS Code (HTML and JS) ★★★★★ FREE	 ng1.5 components uti Ireneusz Patalas ⬇ 23.4K Angular 1.5 components utility ★★★★★ FREE

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IntelliSense is a general term for various code editing features including: code completion, parameter info, quick info, and member lists. IntelliSense features are sometimes called by other names such as "code completion", "content assist", and "code hinting."

```
1  const express = require('express')
2  const app = express()
3
4  app.
```



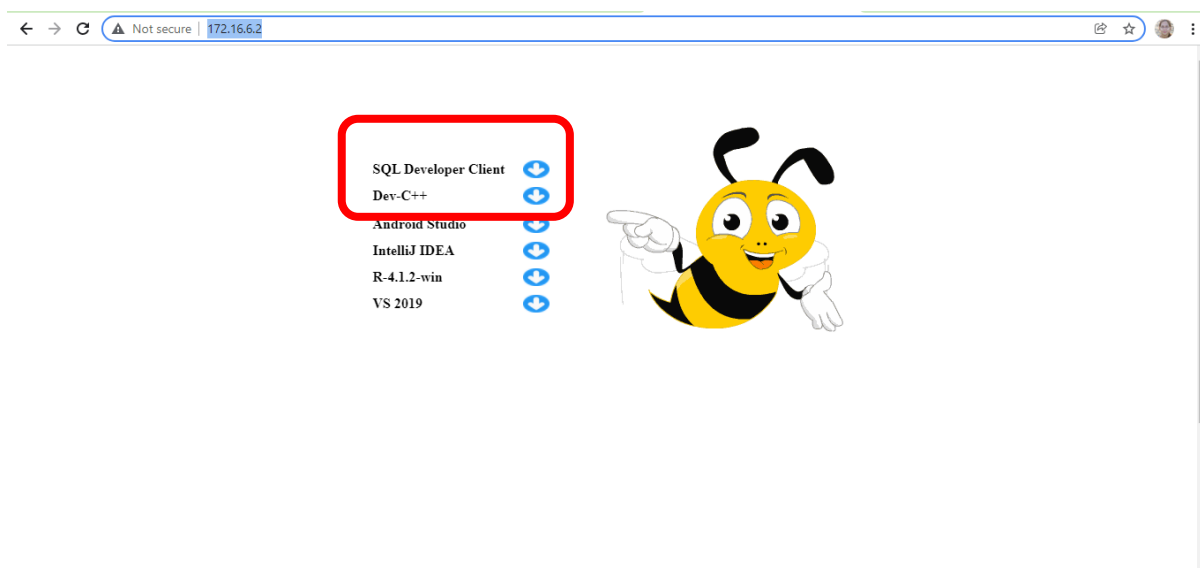
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Objectives

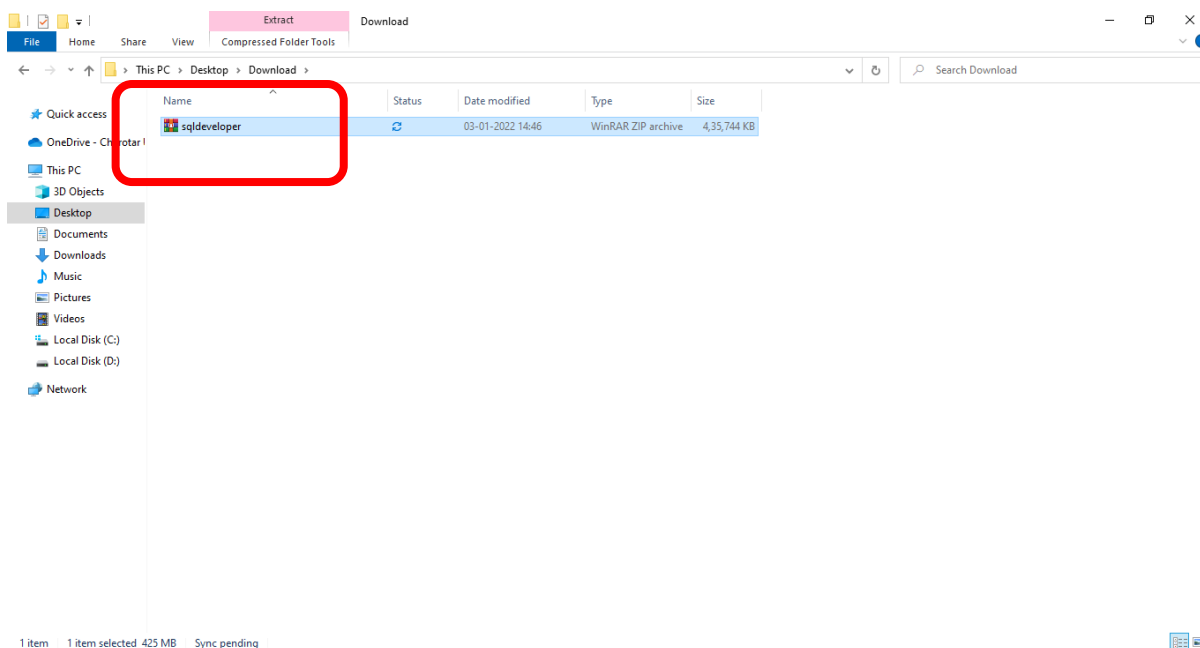
- To make students acquaint with installation of Database Client Software.

Step: 1 Go to <http://172.16.6.2/>

Download SQL Developer Client

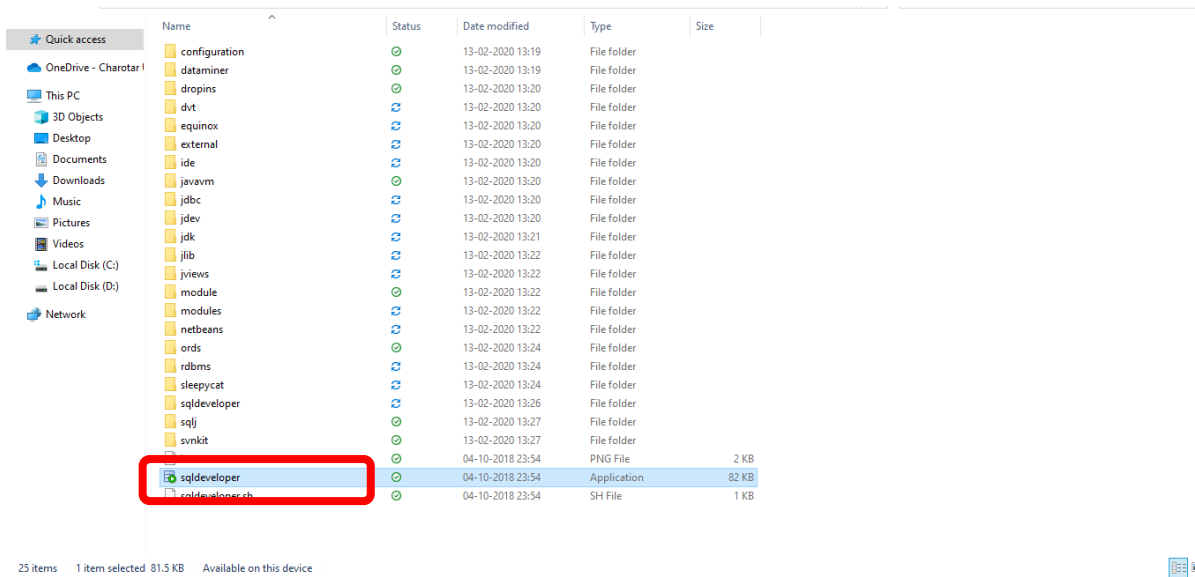


This will download the zip with name **sqldeveloper**. Unzip it.

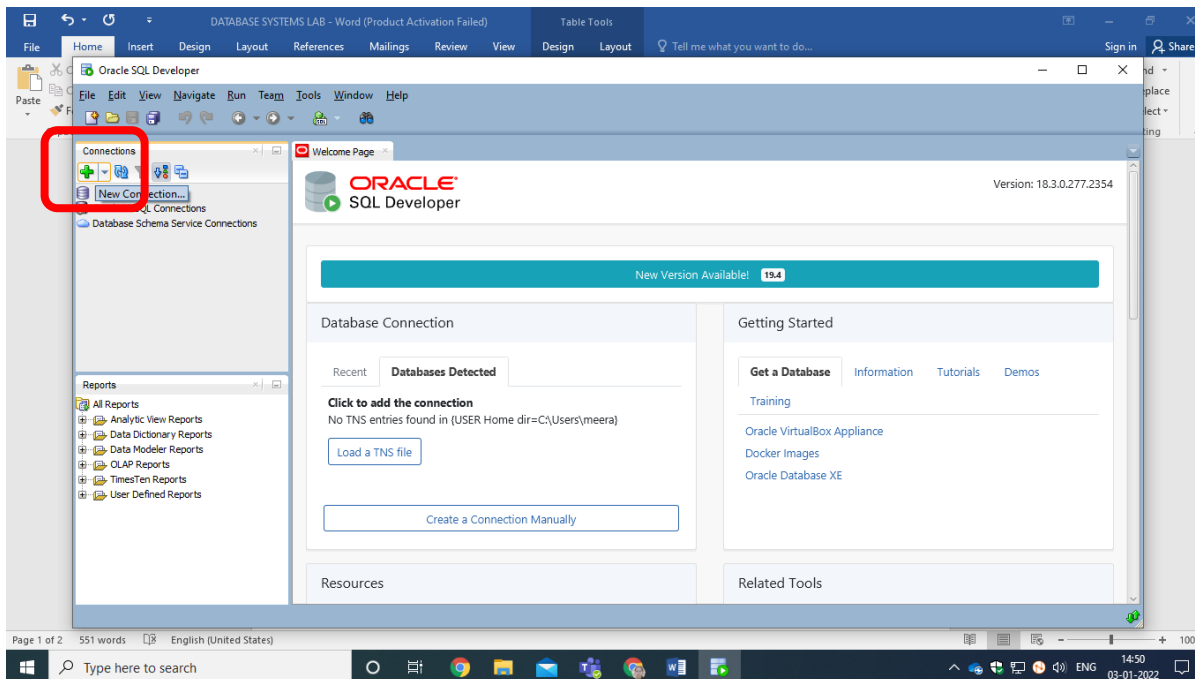


Step: 2 Double click on **sqldeveloper** Application

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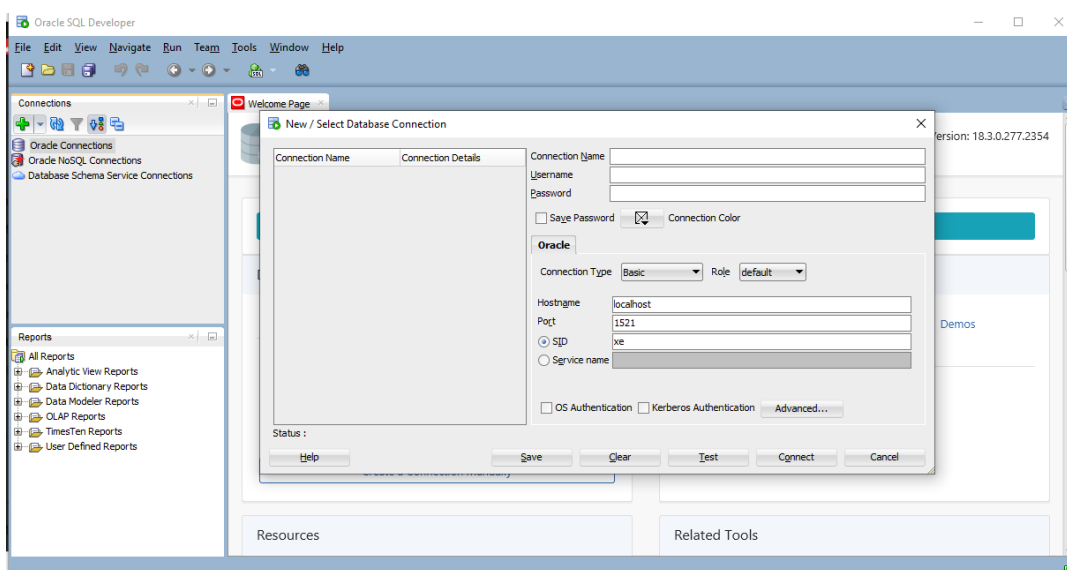


It will open the Oracle SQL Developer.



Step: 3 Click on  Sign, the following screen will appear.

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Step: 4 Give Proper Connection Name for example CMPICA

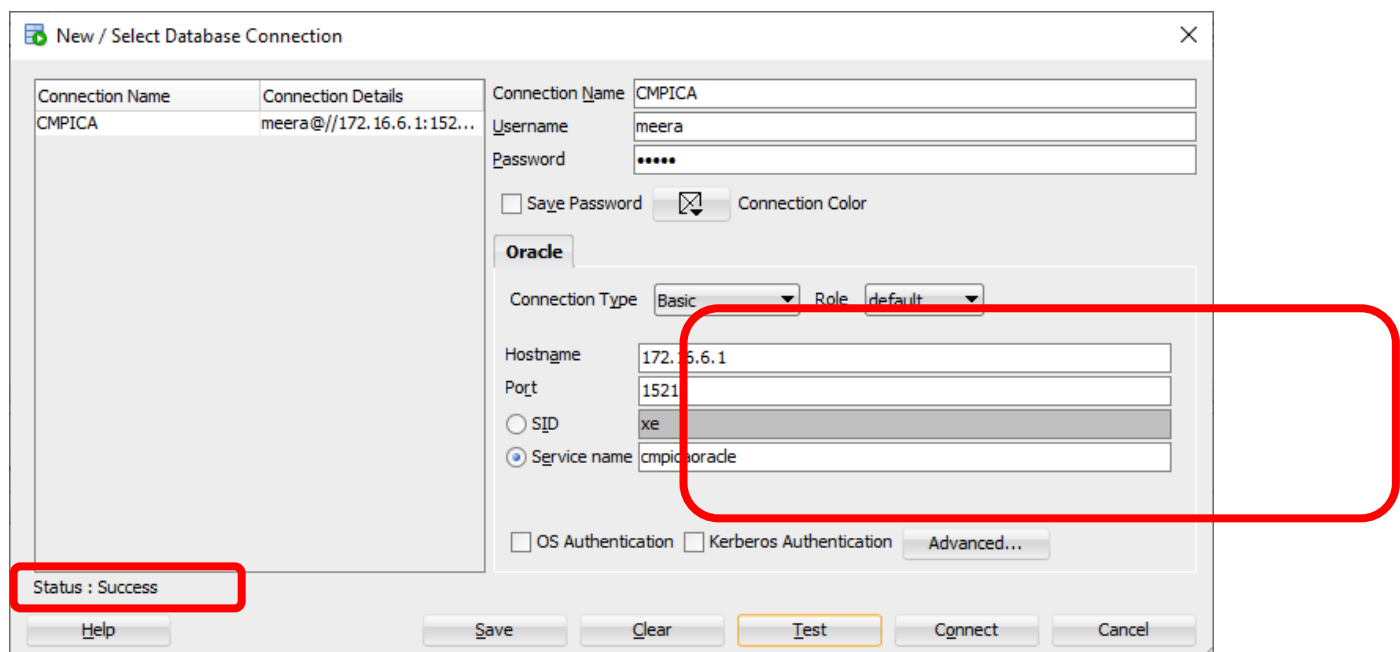
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

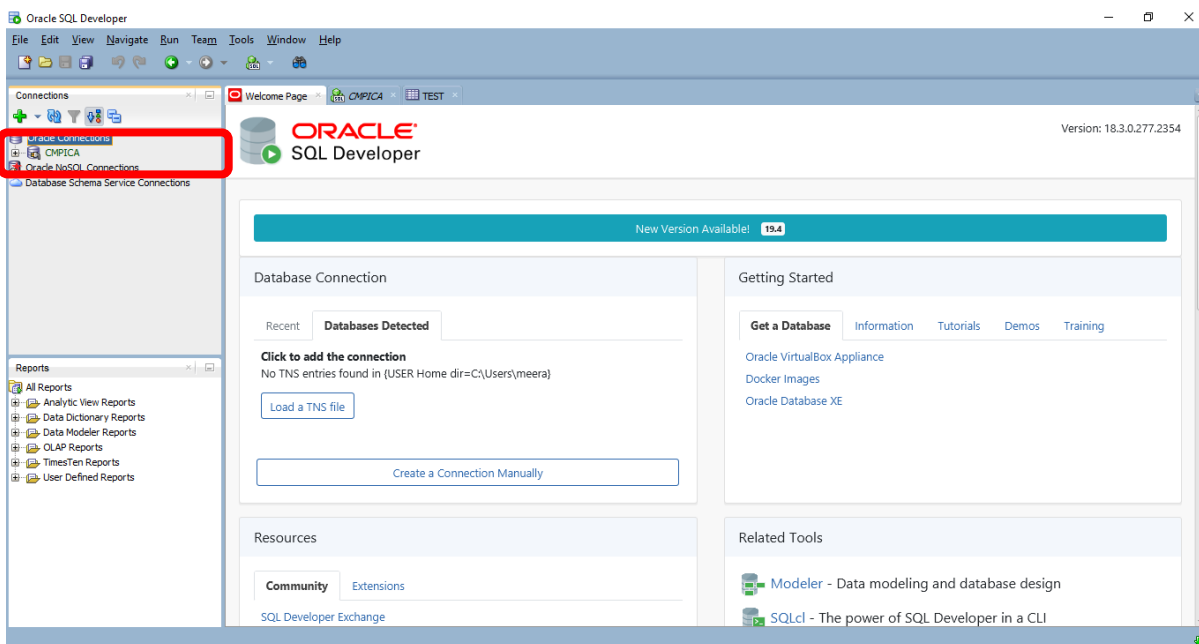
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **success** should get displayed.

Now click on Connect Button.

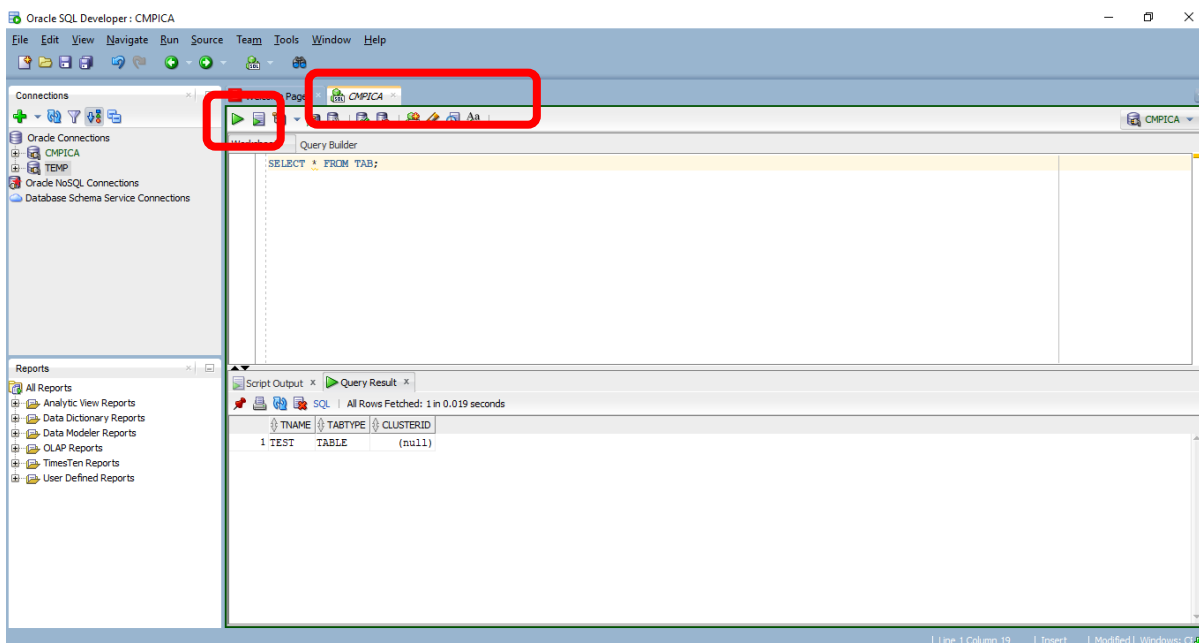


Your newly created connection should get displayed on the left panel.

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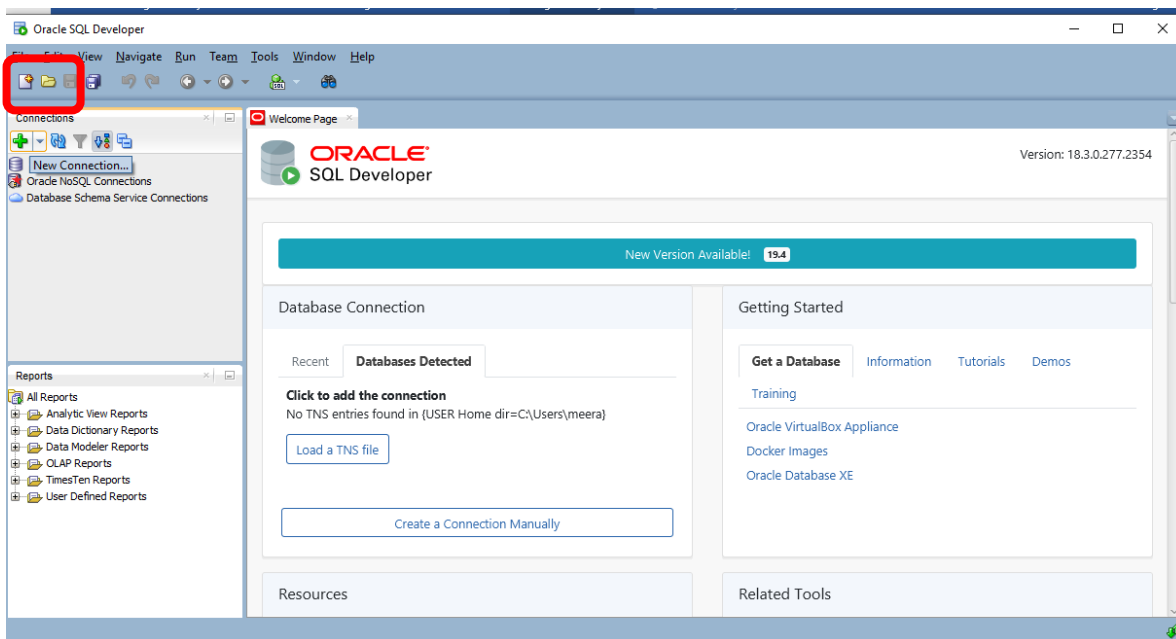
Step 5 On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter.



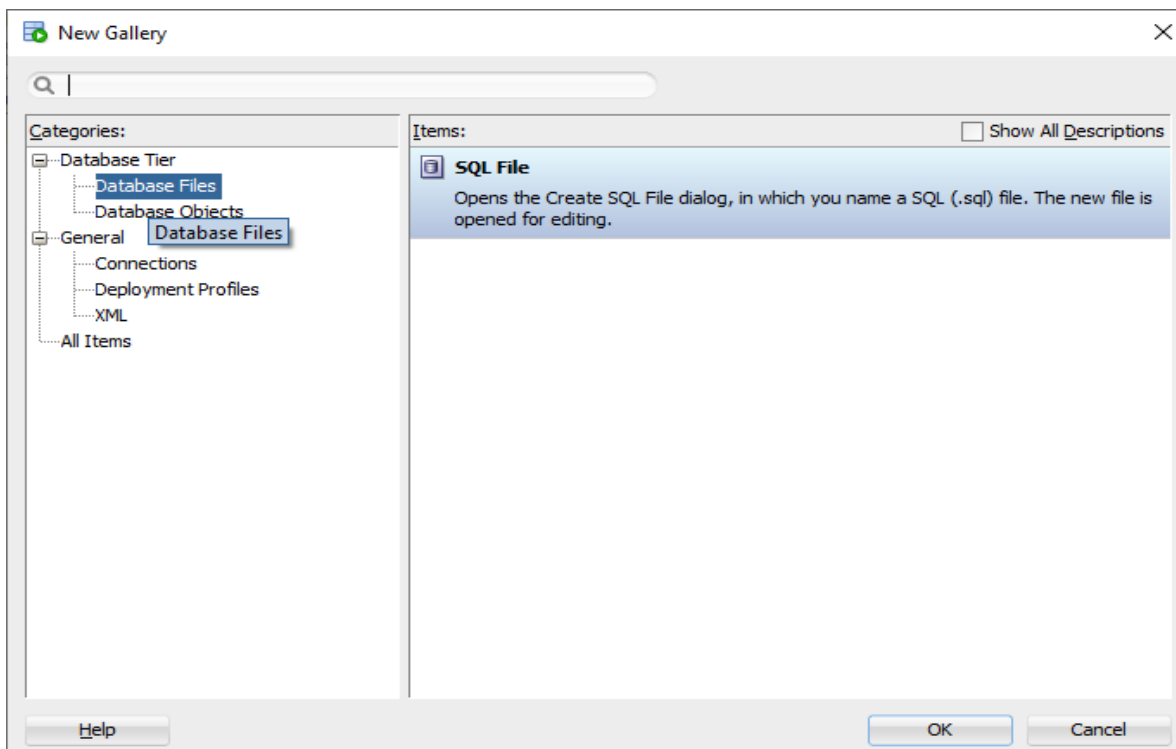
Step 6 IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

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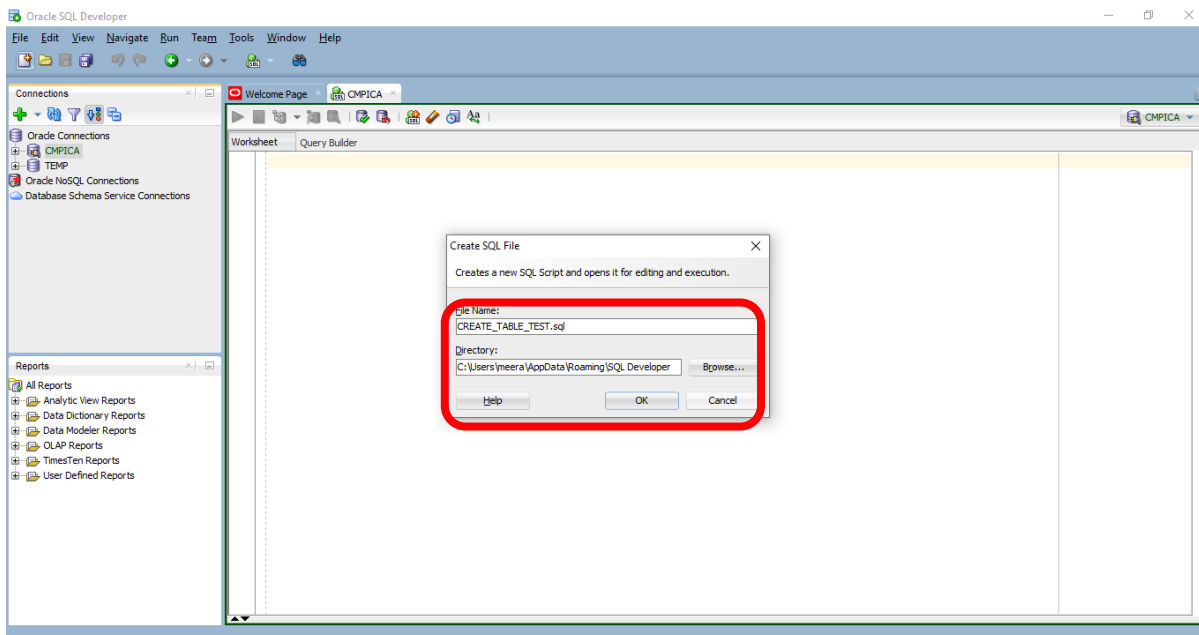


The following screen will appear.



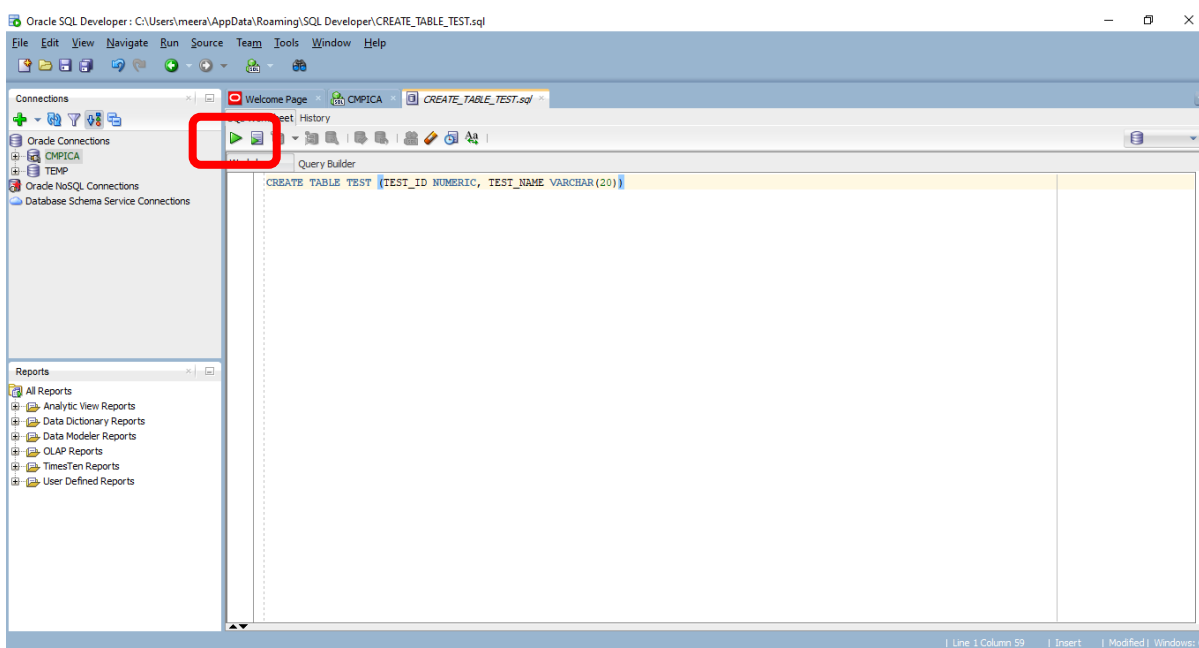
Give proper file name and save to the proper location. **[IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]**

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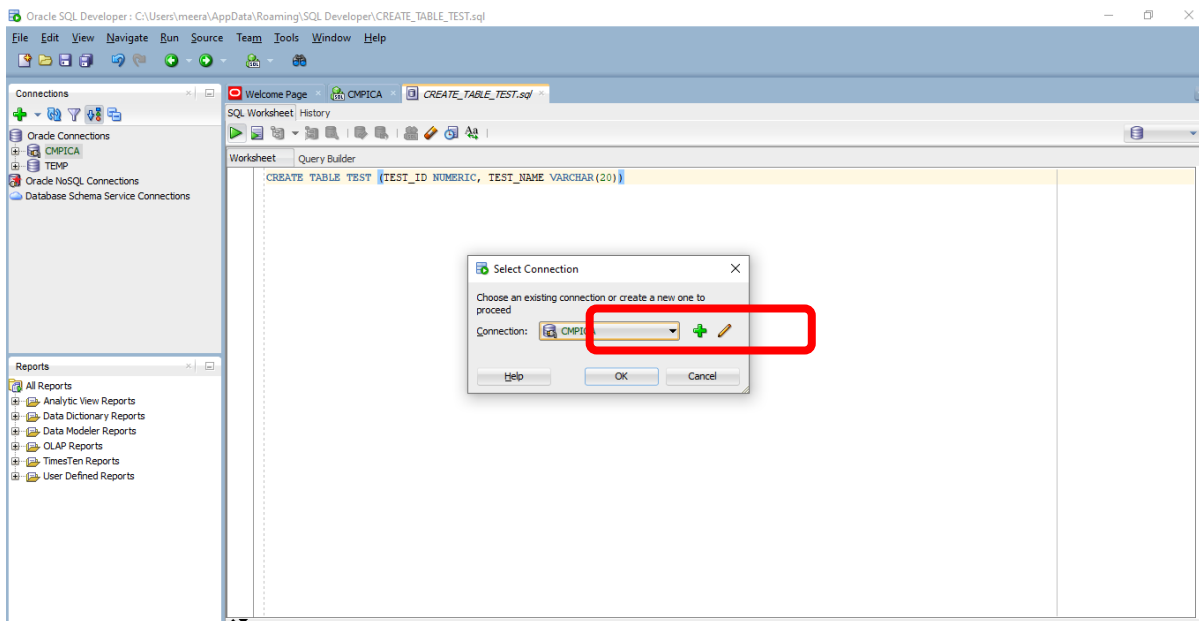
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

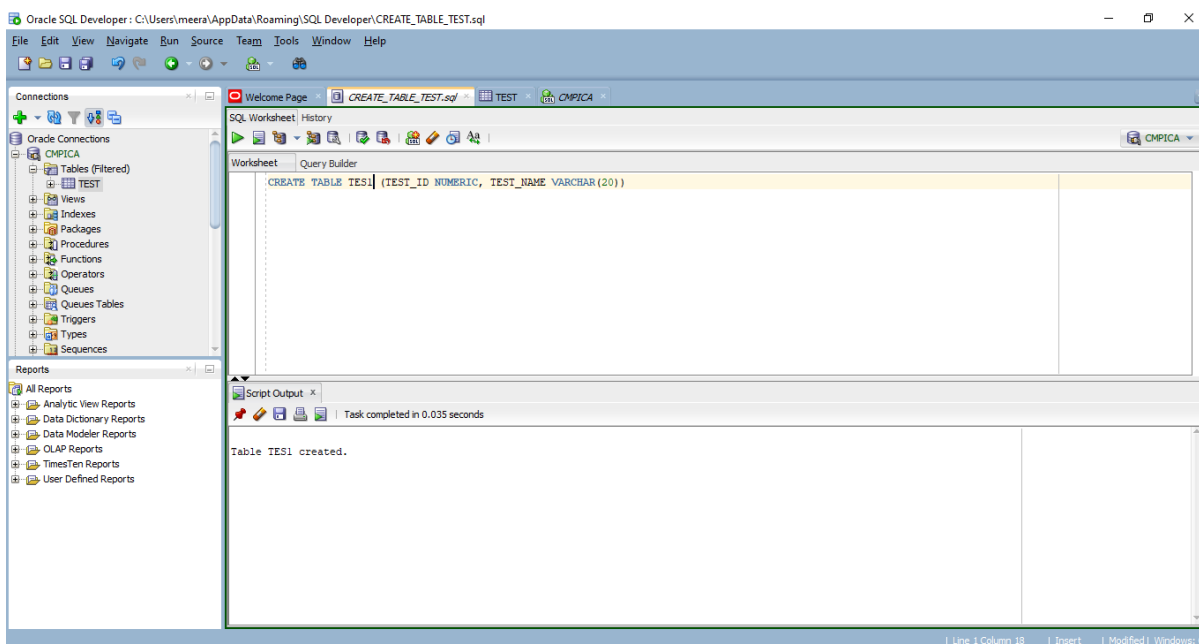


Select the appropriate connection and press OK.

Tool Based Manual || BSc-IT || SEM IV || CA507 : Advanced Database System



Observe execution result in bottom panel.



Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.

Happy Database Exploration!!!



Prepared by : Ms. Hema Patel

Id: 1032

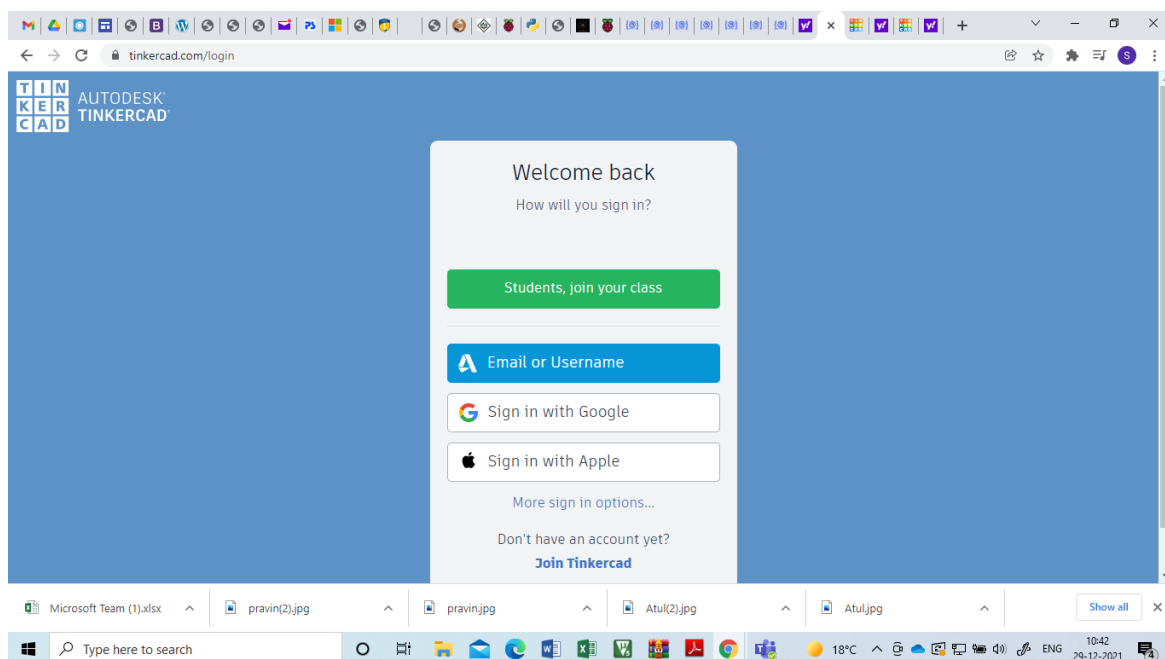
Contents

Tinkercad Previw	2
Creating Circuit using Timkercad.....	2
Preview of armKEIL	5
Creating a project using KEIL.....	6

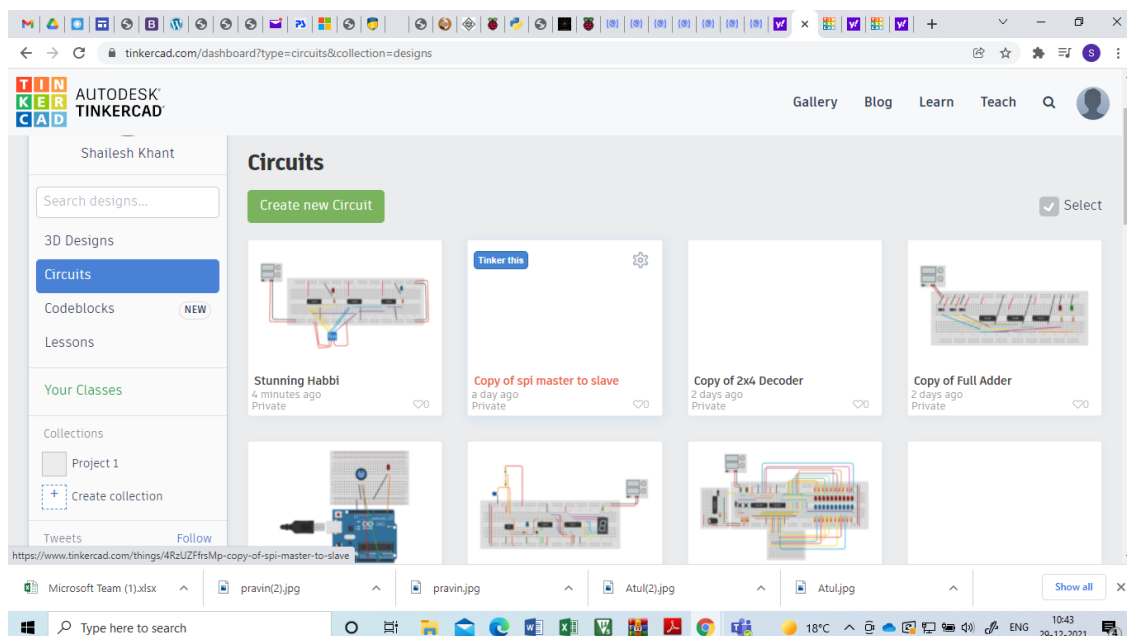
Simulation Using Tinker cad Tool.

Getting Started with Tinkercad

1. Open www.tinkercad.com
2. Login using Google or Facebook account.

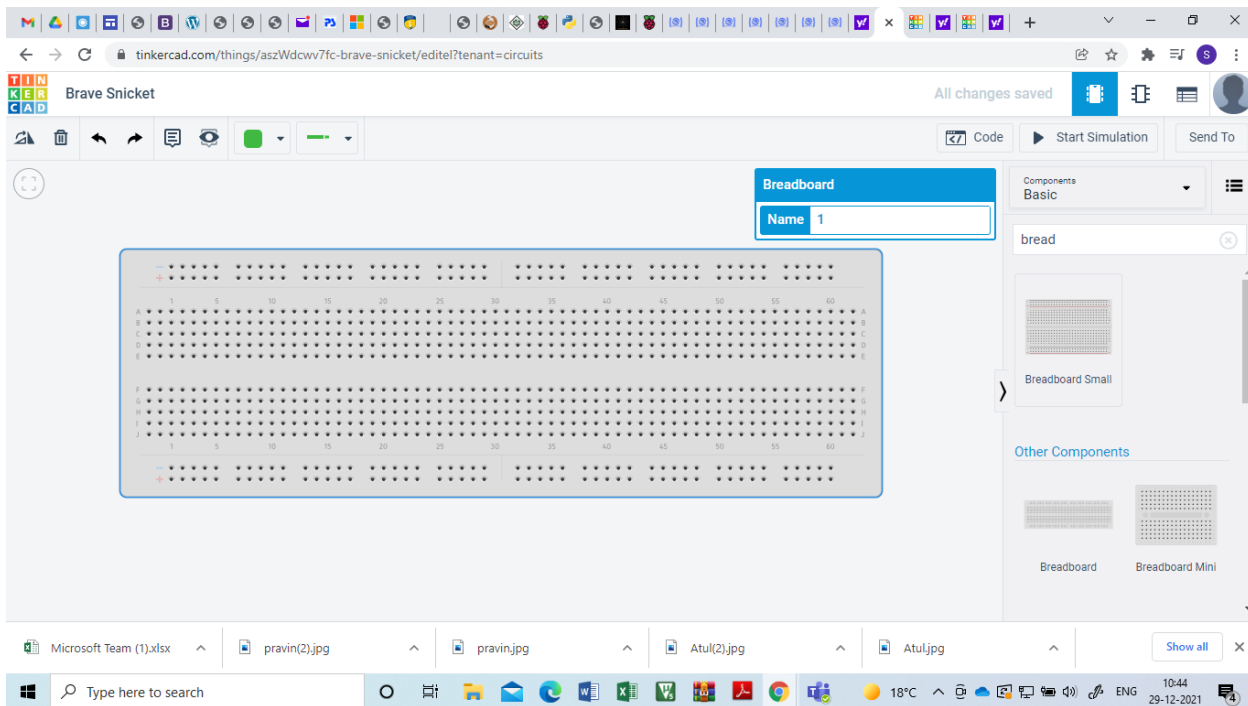


3. Click on new circuit to design new circuit based on electronics hardware.

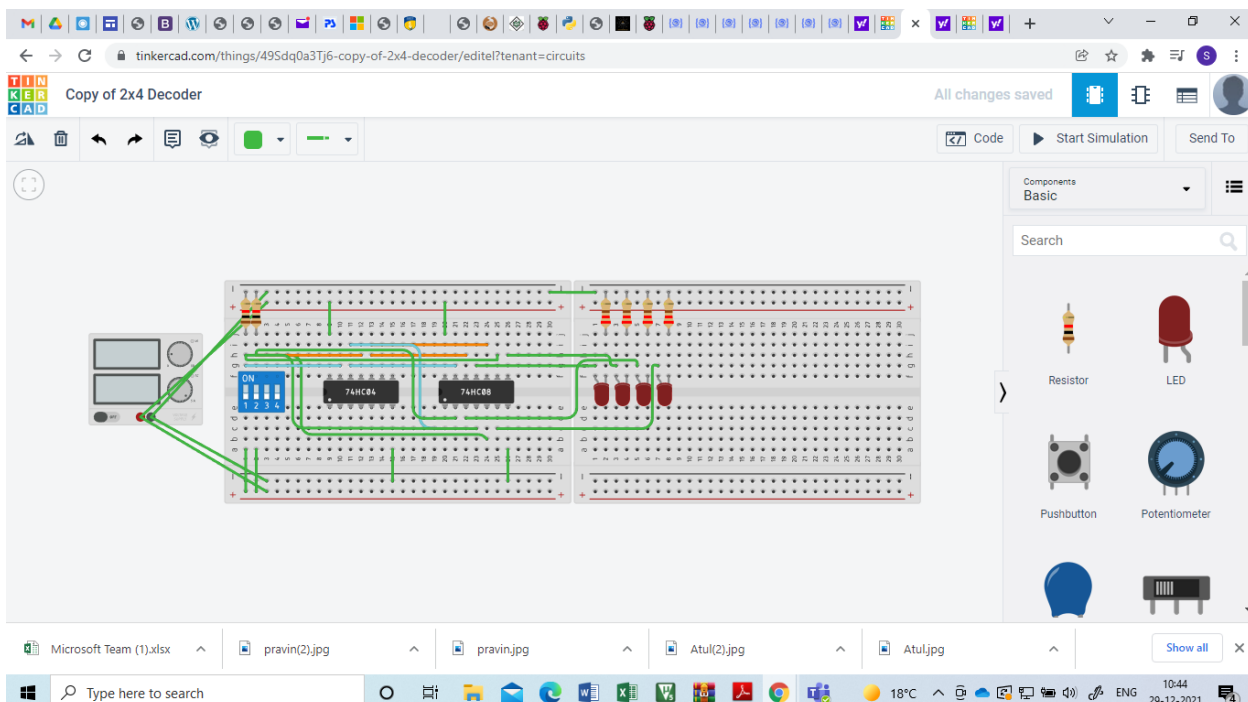


Tool Based Manual || B. Sc. (IT) || SEM IV || CA512 : Fundamentals of Embedded Programming

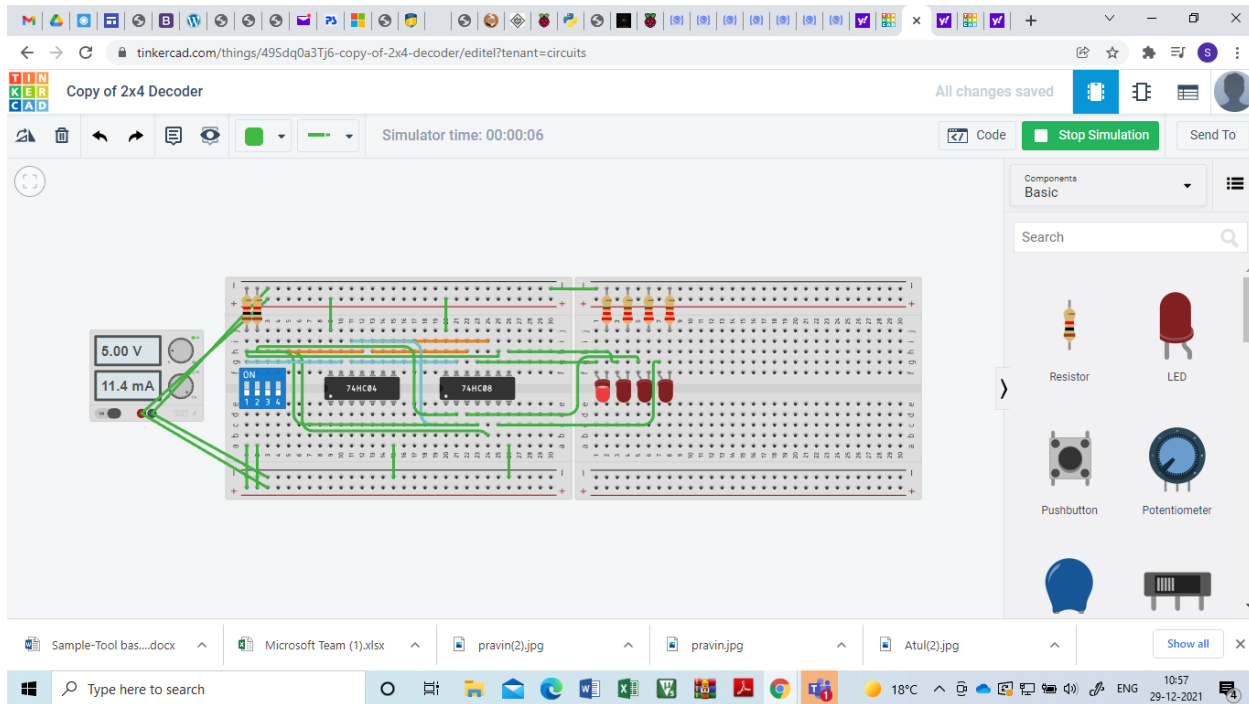
4. Once new circuit is opened then find out the desired components and placed on board.



5. Design whole circuit and check out proper connections.

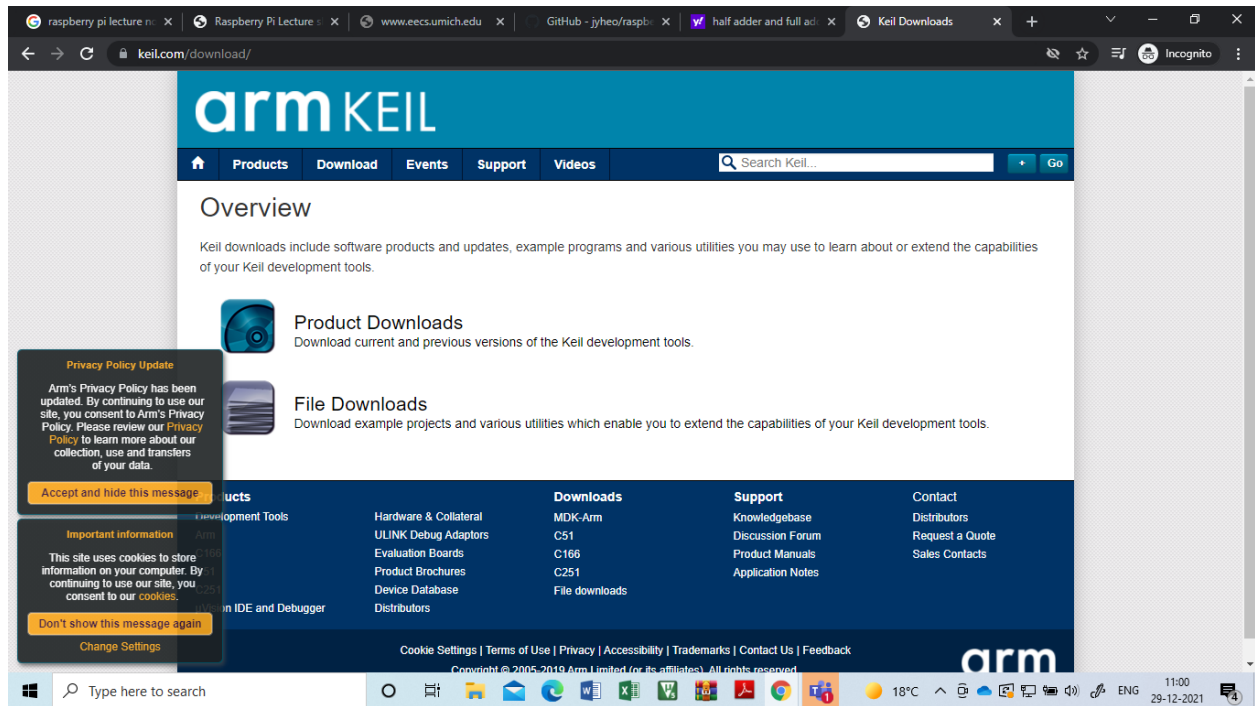


6. Click on start simulation to check the output.

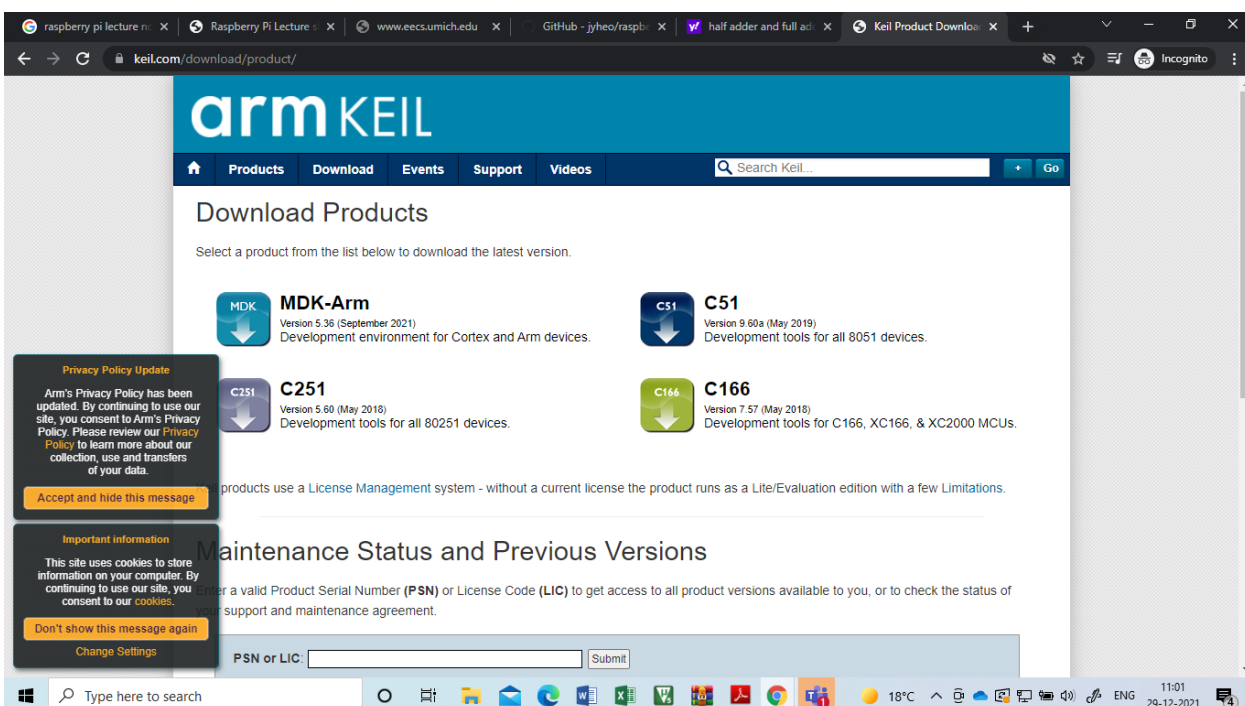


Simulation using Keil Software

1. Open www.Keil.com/downloads

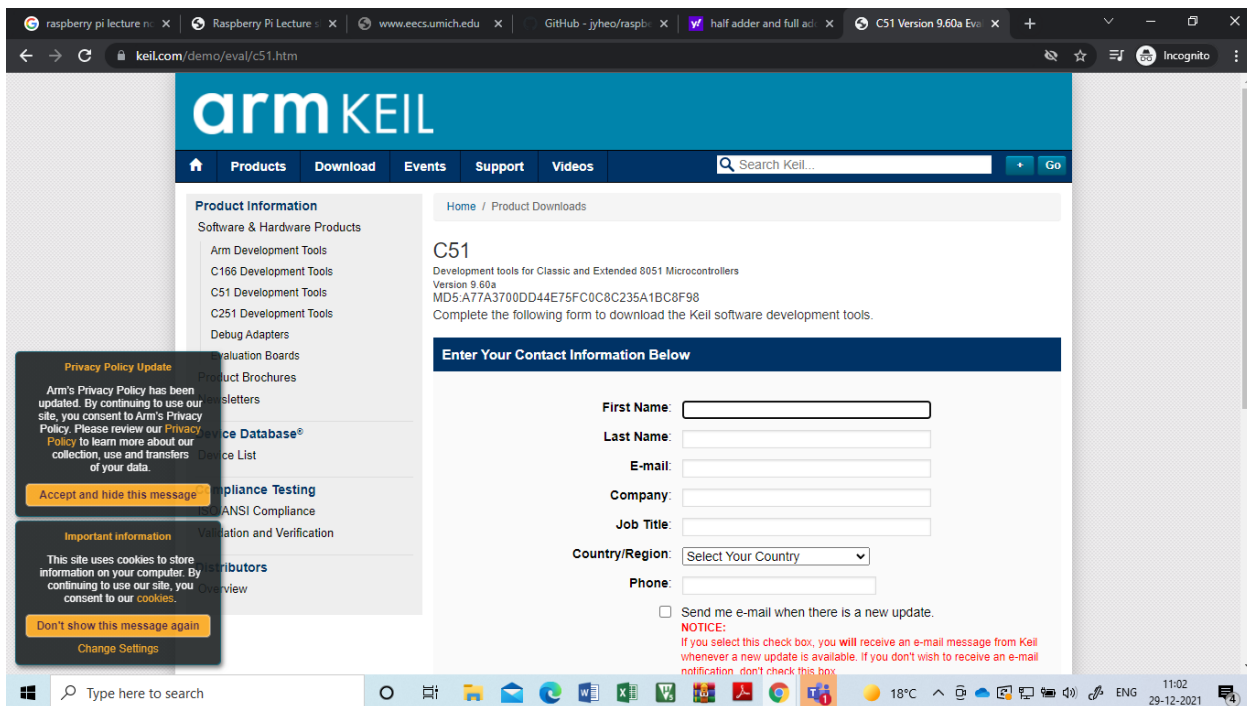


2. Go to product downloads and select C51 tool for download.

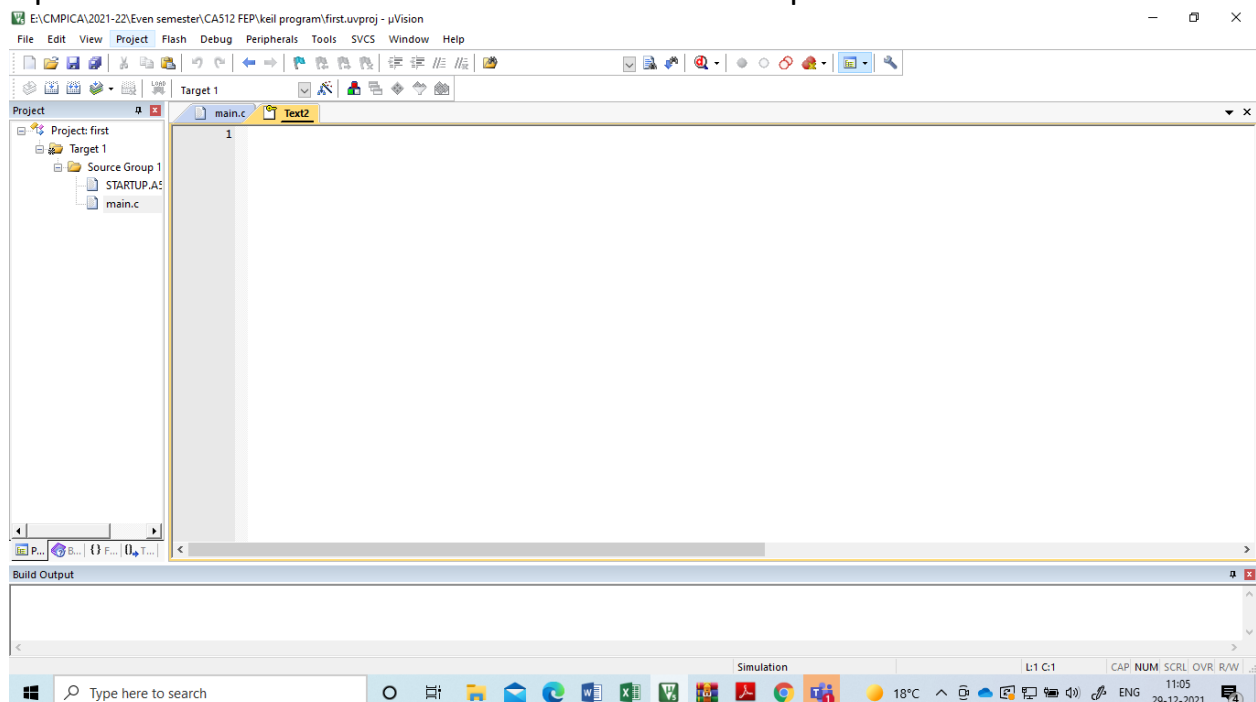


3. Fill your details.

Tool Based Manual || B. Sc. (IT) || SEM IV || CA512 : Fundamentals of Embedded Programming

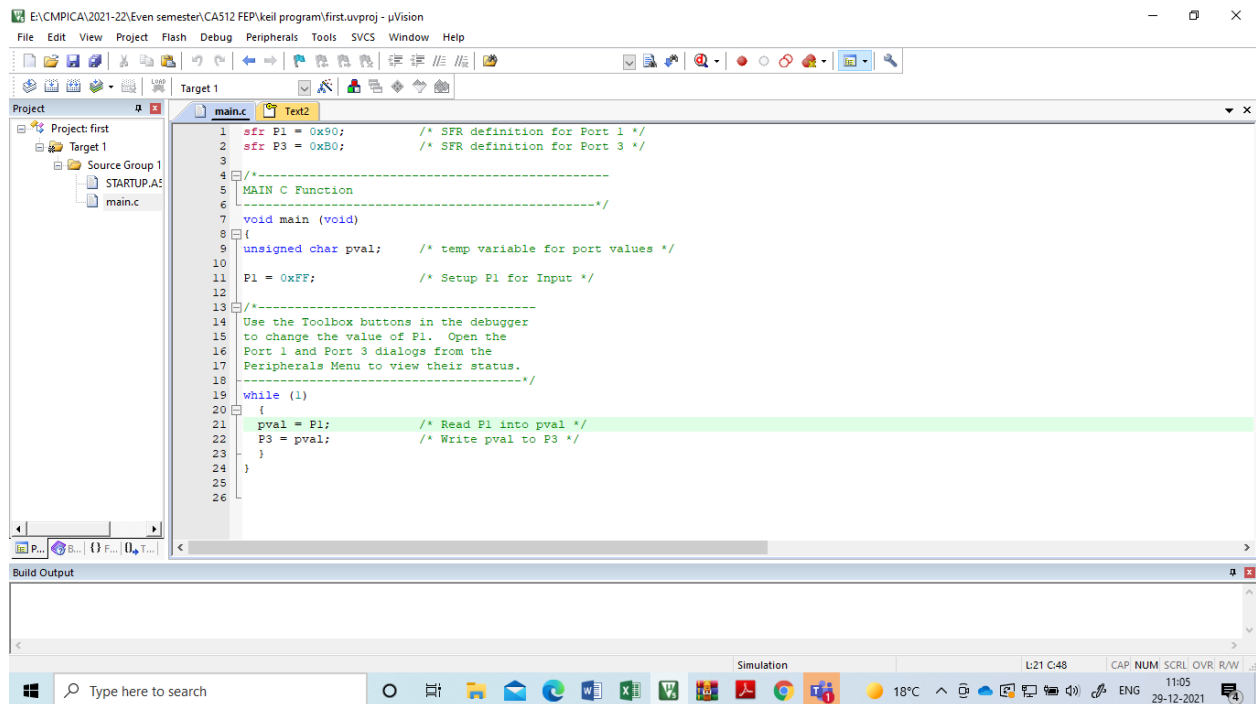


4. Download the keil version C51.
5. Open Keil software from start menu or from desktop icon.

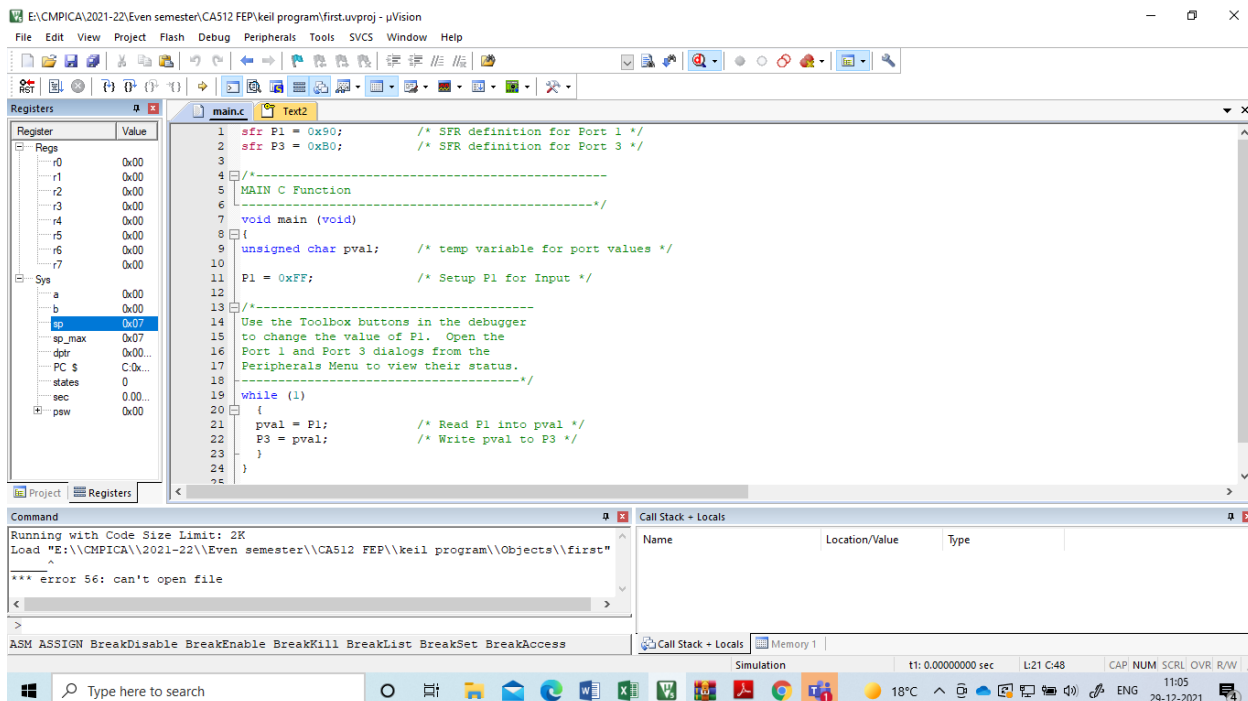


6. Create new project and create new file in that.

Tool Based Manual || B. Sc. (IT) || SEM IV || CA512 : Fundamentals of Embedded Programming



7. Start the debug session.



8. Check out various registers to see the outputs.